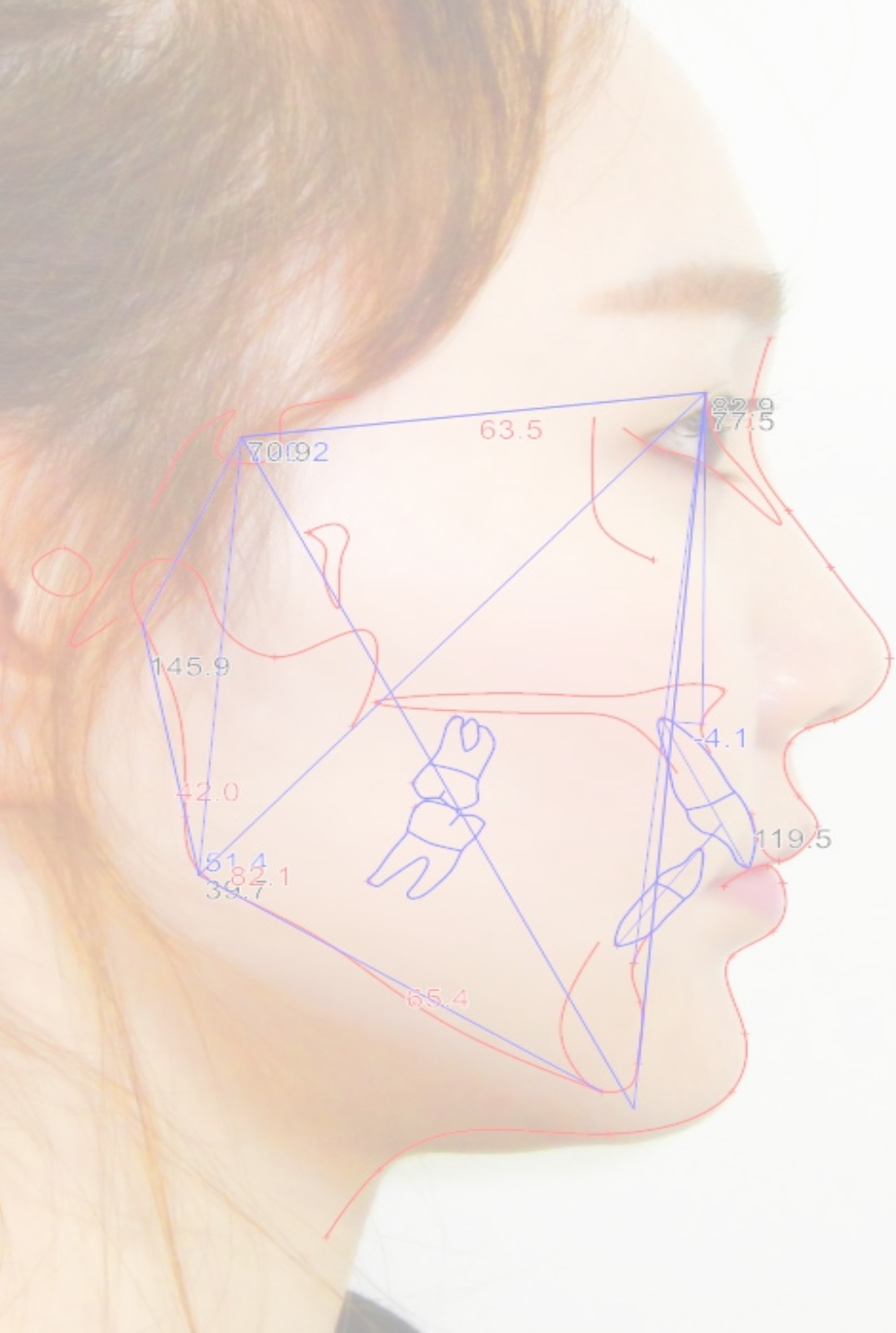
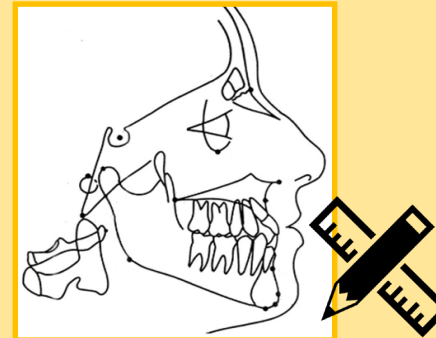




WillCeph User's Guide



HDX WILL North America
Education & Development dept.

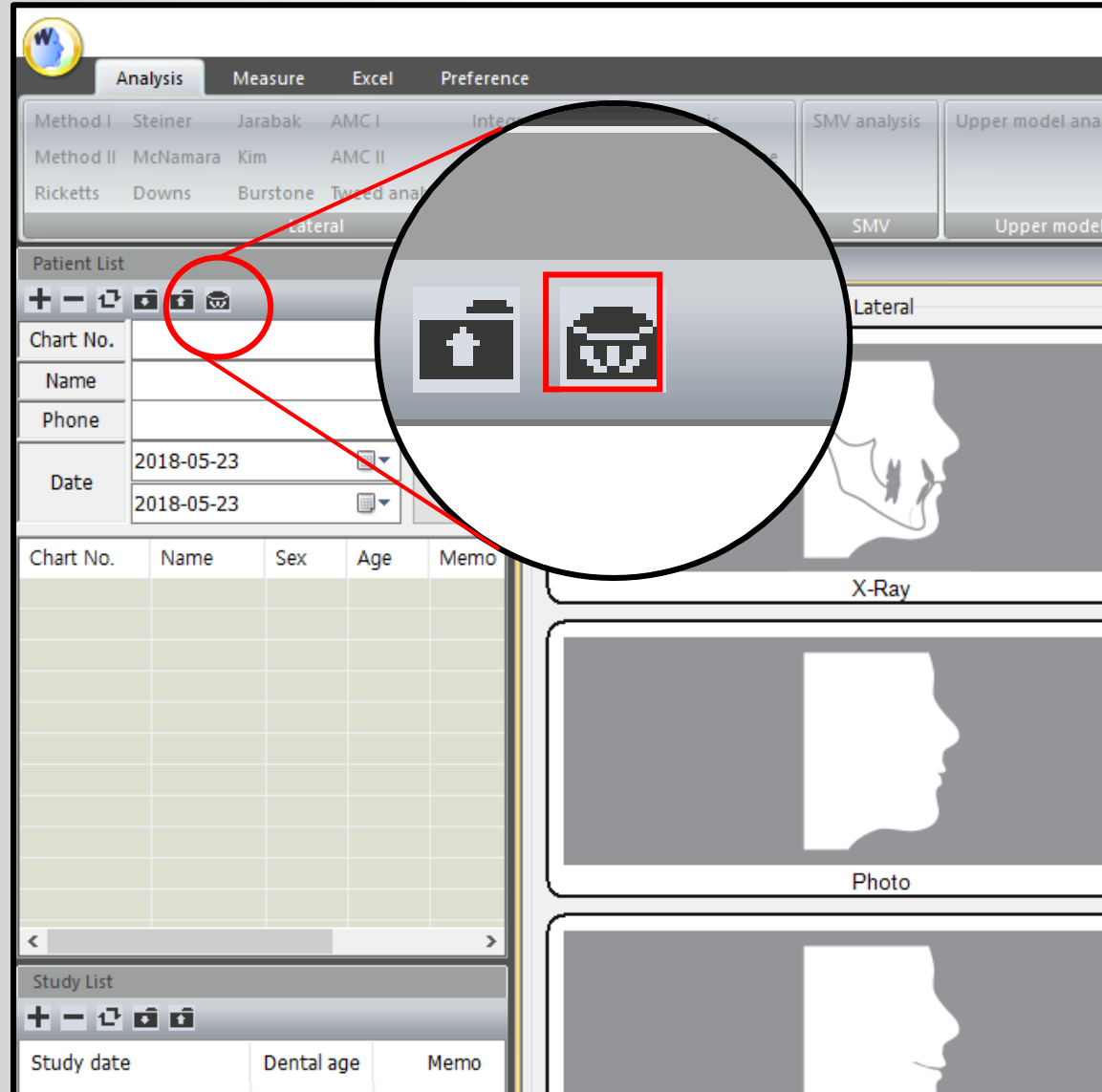


3. Analyzing and Utilizing

Patient Registration

- 1) Import images from Will-Master database
- 2) Manually import Images

1) Import images from Will-Master



Import data from Will-Master

Dentri α images are calibrated, no additional calibration steps necessary for cephalometric analysis after images are imported from the data-base.

- ① Click  button under patient list

1) Import images from Will-Master

[illegible]

Step 1 Will Master

: Select patient and images for analysis

- ① Search patient by chart number or patient name then click on search
- ② Additional search features
 - Set the exact date by clicking on the downward arrow

Week Month Today

- Set the exact date by clicking on the downward arrow

May, 2018						
Sun	Mon	Tue	Wed	Thu	Fri	Sat
29	30	1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31	1	2
3	4	5	6	7	8	9

Today: 5/23/2018

1) Import images from Will-Master

WillCommunication

Step 1 : Will-Master | Step 2 : WillCeph | Step 3 : Set category

Step 1 : First Select the patient and image for

③

Patient List

Chart Number Search

Patient Name

Today Week Month

☐ 2018-05-23 ~ 2018-05-23

Chart No.	Name	Sex	Age	Bir
777777	WILLCEPH	M	28	19

④

Image List

Chart No.	Category
777777	PANORAMA
777777	CEPHALO[LATERAL]
777777	CEPHALO[FRONTAL]
777777	CAMERA[LEFT]
777777	CAMERA[CENTER]

Preview image

< Prev **⑤** Next >

③ Click on patient name in the grid

➤ (name will highlight in blue)

④ From **Image List**, select desired images
(Hold **ctrl** key to select multiple images)

⑤ Click "Next"

1) Import images from Will-Master

WillCommunication

Step 1 : Will-Master | Step 2 : WillCeph | Step 3 : Set category

Step 2 : Select the study which to copy the

Patient List

① Add selected Will-Master patient

Chart Number: 777777 New

Patient Name: WILLCEPH Search

Today Week Month

☐ 2018-05-23 ~ 2018-05-23

Chart No.	Name	Sex	Age
777777	WILLCEPH	M	28

< >

< Prev Next >

Step 2 Will Ceph

: Enter study details

- ① For new patient, click **Add selected Will-Master patient**
- Otherwise, the patient's name will auto populate

1) Import images from Will-Master

WillCommunication

Step 1 : Will-Master | Step 2 : WillCeph | Step 3 : Set category

Step 2 : Select the study which to copy the

Patient List

Add selected Will-Master patient

Chart Number: 777777 New

Patient Name: WILLCEPH Search

Today Week Month

☐ 2018-05-23 ~ 2018-05-23

Chart No.	Name	Sex	Age
777777	WILLCEPH	M	28

Add study

New

Study date	Dental age
[1.PRE-TREATMENT]	
2018-05-23 15:40:15	AUDLT

< Prev

Next >

② Select patient (name will be highlighted in blue)

③ Click "New"

④ Enter study information (fill in details)

Add study

Detail

Dental age: IA

Status: PRE-TREATMENT

Memo

Description

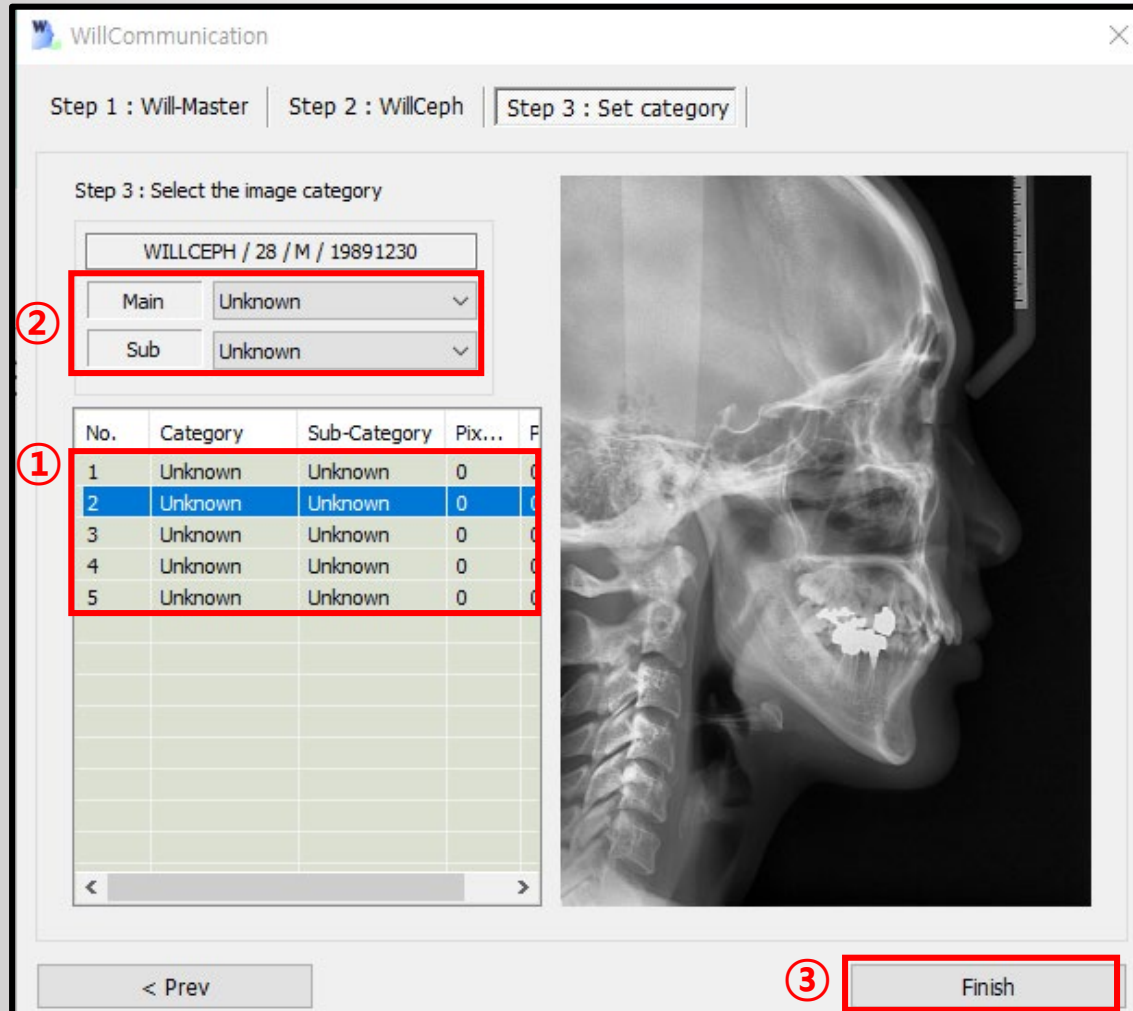
Init. Add

IA
IA
IC
IIA : Completion of Primary occlusion
IIC : Eruption phase of permanent first molar or incisors
IIIA : Eruption of permanent first molar or incisors completed
IIIB : Exchange phase of lateral teeth
IIIC : Eruption phase of permanent second molar
IV A : Eruption of permanent second molar completed
IV C : Eruption phase of permanent third molar
VA : Eruption of permanent second molar completed
ADULT

PRE-TREATMENT
PRE-TREATMENT
MID-TREATMENT
POST-TREATMENT
RETENTION

⑤ Select the created study, click "Next"

1) Import images from Will-Master



WillCommunication

Step 1 : Will-Master | Step 2 : WillCeph | Step 3 : Set category

Step 3 : Select the image category

WILLCEPH / 28 / M / 19891230

Main: Unknown
Sub: Unknown

No.	Category	Sub-Category	Pix...	P
1	Unknown	Unknown	0	0
2	Unknown	Unknown	0	0
3	Unknown	Unknown	0	0
4	Unknown	Unknown	0	0
5	Unknown	Unknown	0	0

Preview image: Lateral cephalogram

< Prev Finish

Step 3 Set Category

: Select the image category

- ① Preview image by selecting the No. category
- ② Assign main and sub categories

Unknown
Unknown
Lateral
Frontal
SMV
Upper model
Lower model
Camera
Panorama
IntraOral
IntraOral-Photo
Etc

Unknown
Unknown
Film
Photo
Smile
Trace

(Example)

	Main	Sub
Lateral Ceph	Lateral	Film
Lateral Photo	Lateral	Photo
Panorama	Panorama	Film

- ③ Click "Finish"

1) Import images from Will-Master

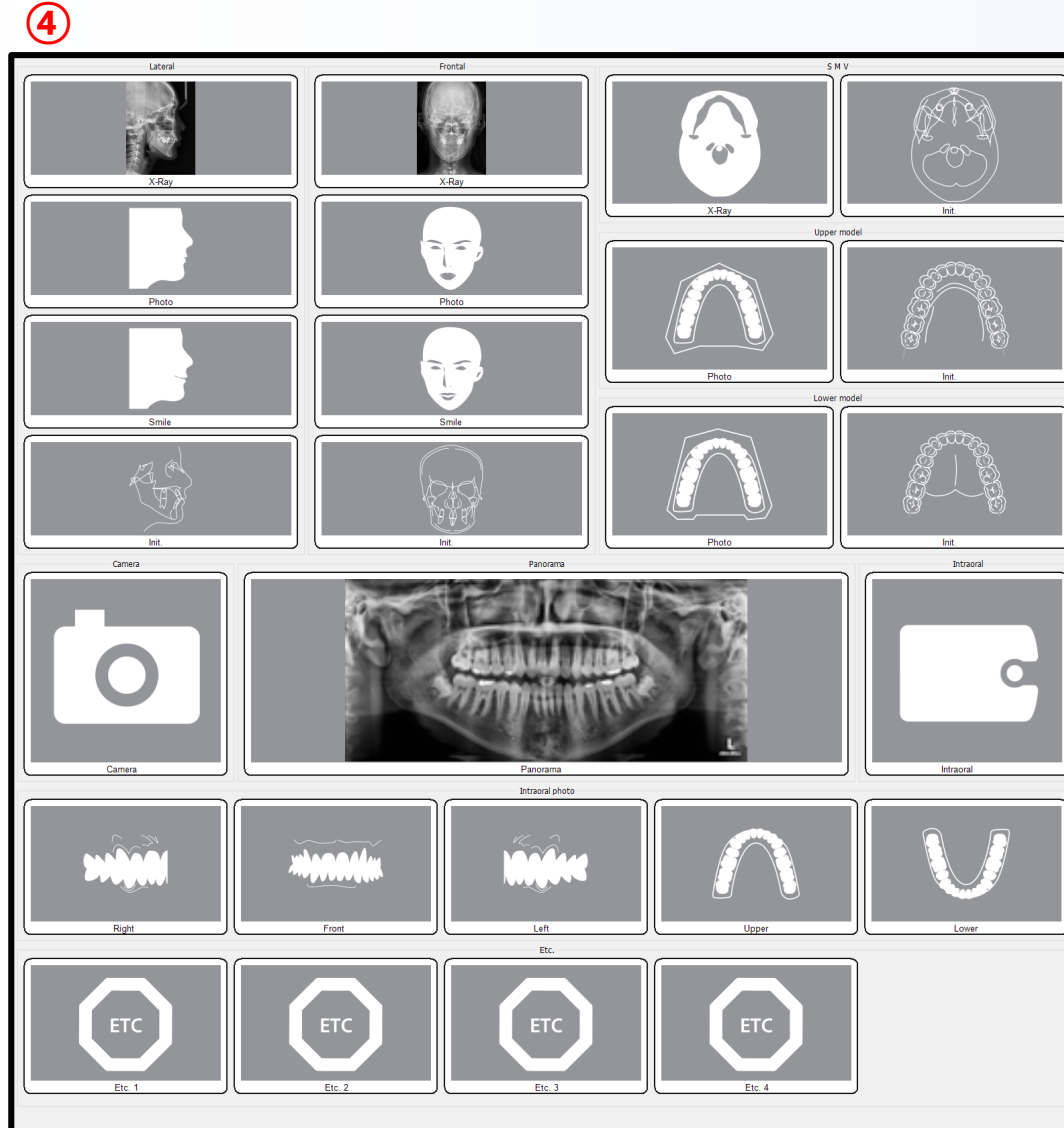
Patient List

Chart No. Today
Name Week
Phone
Date Search

Chart No.	Name	Sex	Age	Memo
777777	WILLCEPH	M	29	

Study List

Study date	Dental age	Memo
[1.PRE-TREATMENT]		
2019-02-07 11:58:45	ADULT	
[2.MID-TREATMENT]		
2018-07-17 11:54:27	ADULT	
[3.POST-TREATMENT]		
2019-02-07 15:30:14	ADULT	



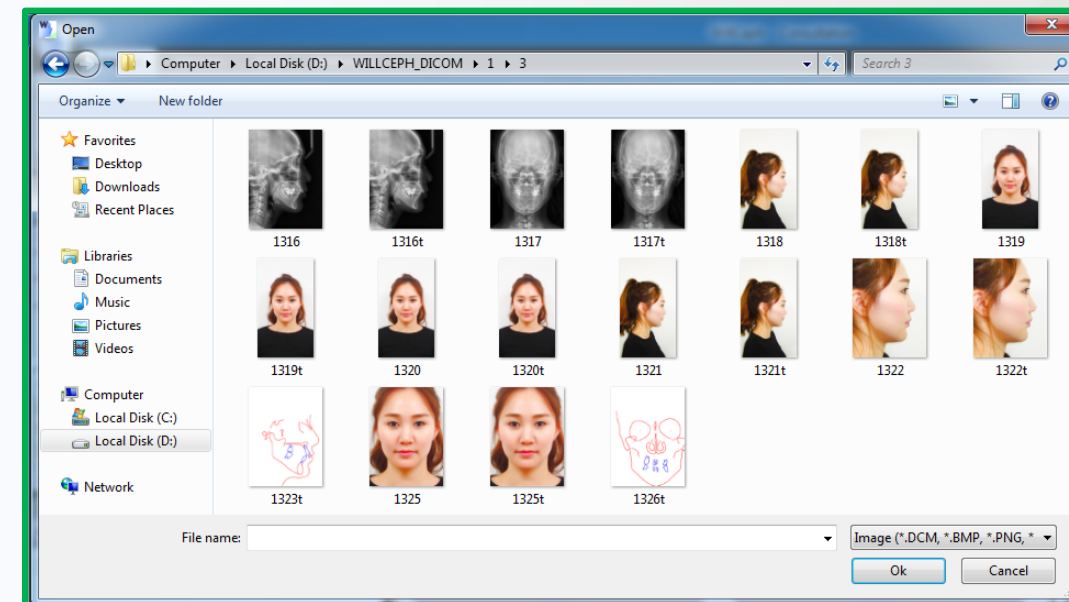
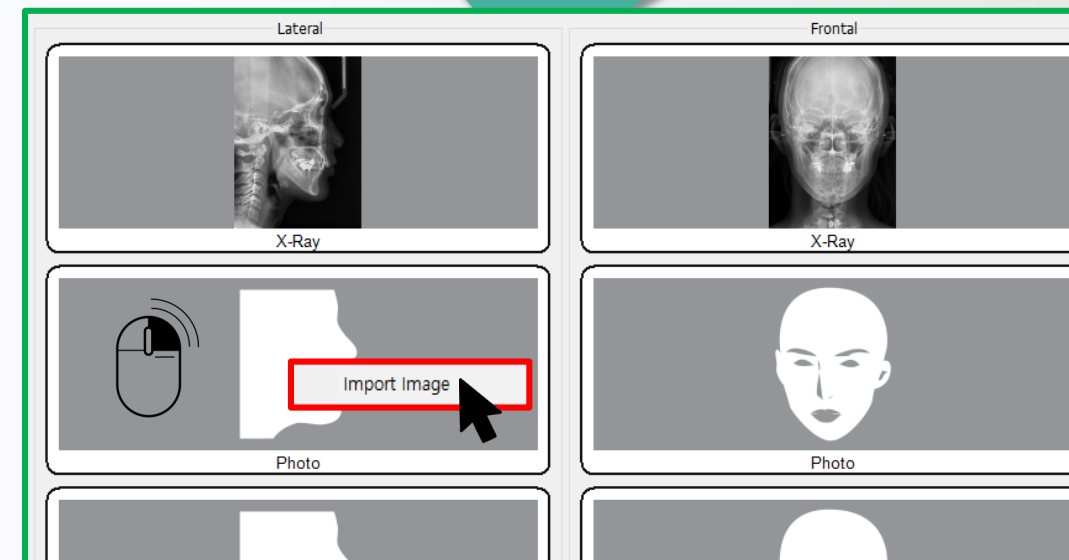
Search Patient

- ① Search registered patient using Search feature
- ② Once the patient is identified by the system patient's information will appear below
- ③ Click on a study to analyze
- ④ Verify the images are in the correct titles

1) Import images from Will-Master

Adding extraoral or intraoral Photo

- ① Right click on a desired spot
- ② Click on "import image" to choose image from folder



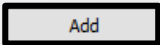
2) Manually import images

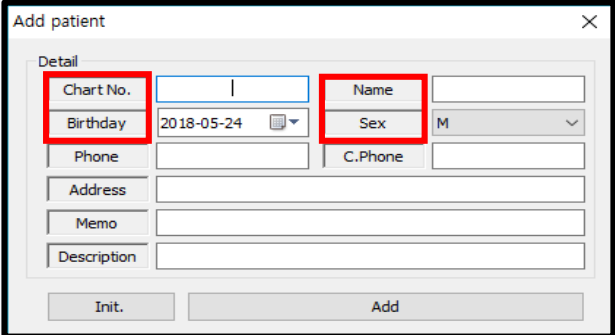
Manual Registration

Images can be added from any imaging system through manual import.

Step 1

: Add patient to the list

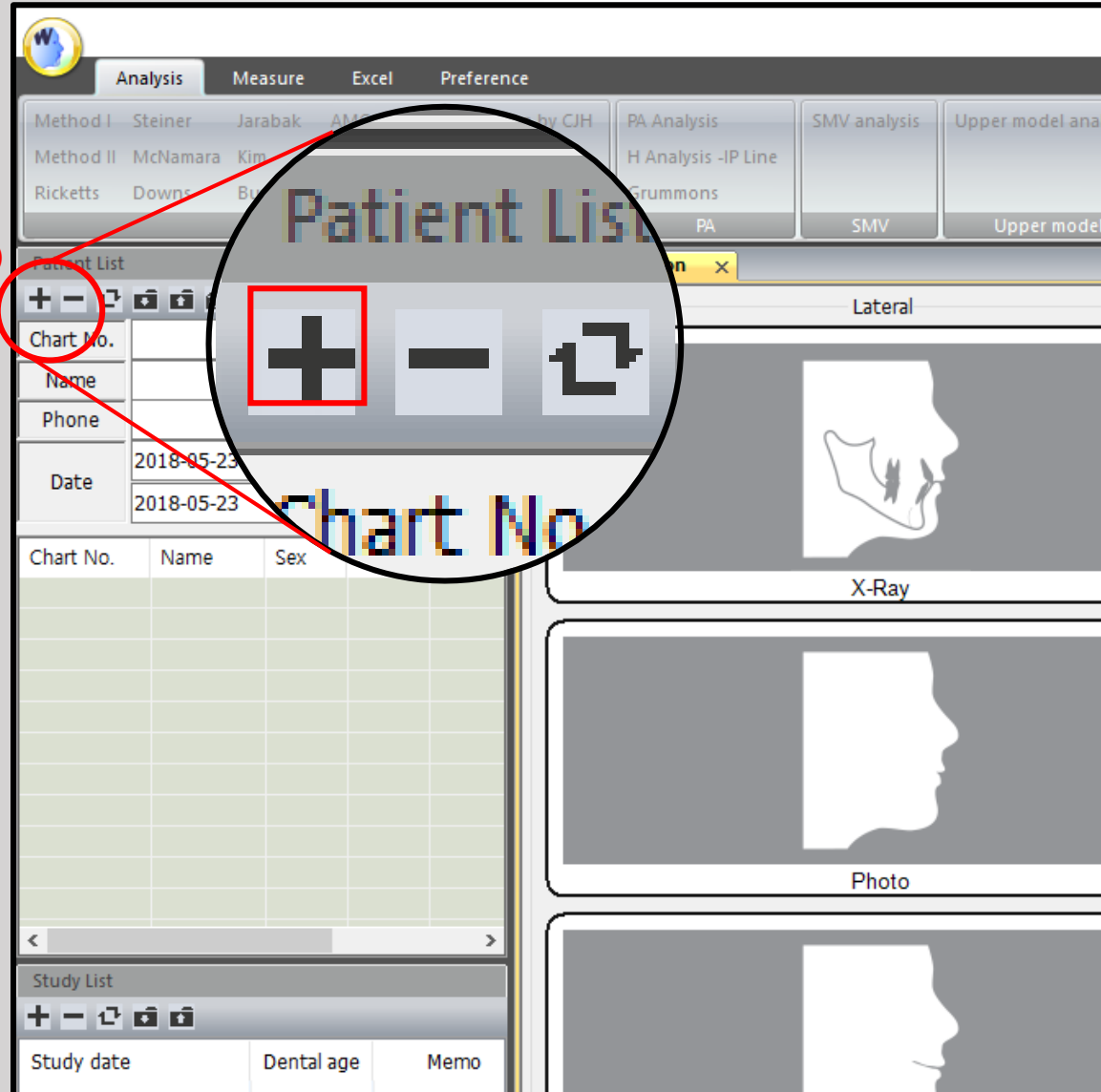
- ① Click  "plus symbol"
- ② Enter patient details then click 
- Required Info: Chart number, Name, DOB, Gender



The 'Add patient' dialog box contains the following fields:

Detail	
Chart No.	
Birthday	2018-05-24
Name	
Sex	M
Phone	
C.Phone	
Address	
Memo	
Description	

Buttons: Init., Add



The 'Patient List' interface shows a table with columns: Chart No., Name, Sex, Date. A red circle highlights the '+', '-', and refresh icons in the top left corner of the table. The table contains two rows of data:

Chart No.	Name	Sex	Date
2018-05-23			
2018-05-23			

Below the table is a 'Study List' section with columns: Study date, Dental age, Memo. It also contains icons for adding, deleting, and refreshing data.

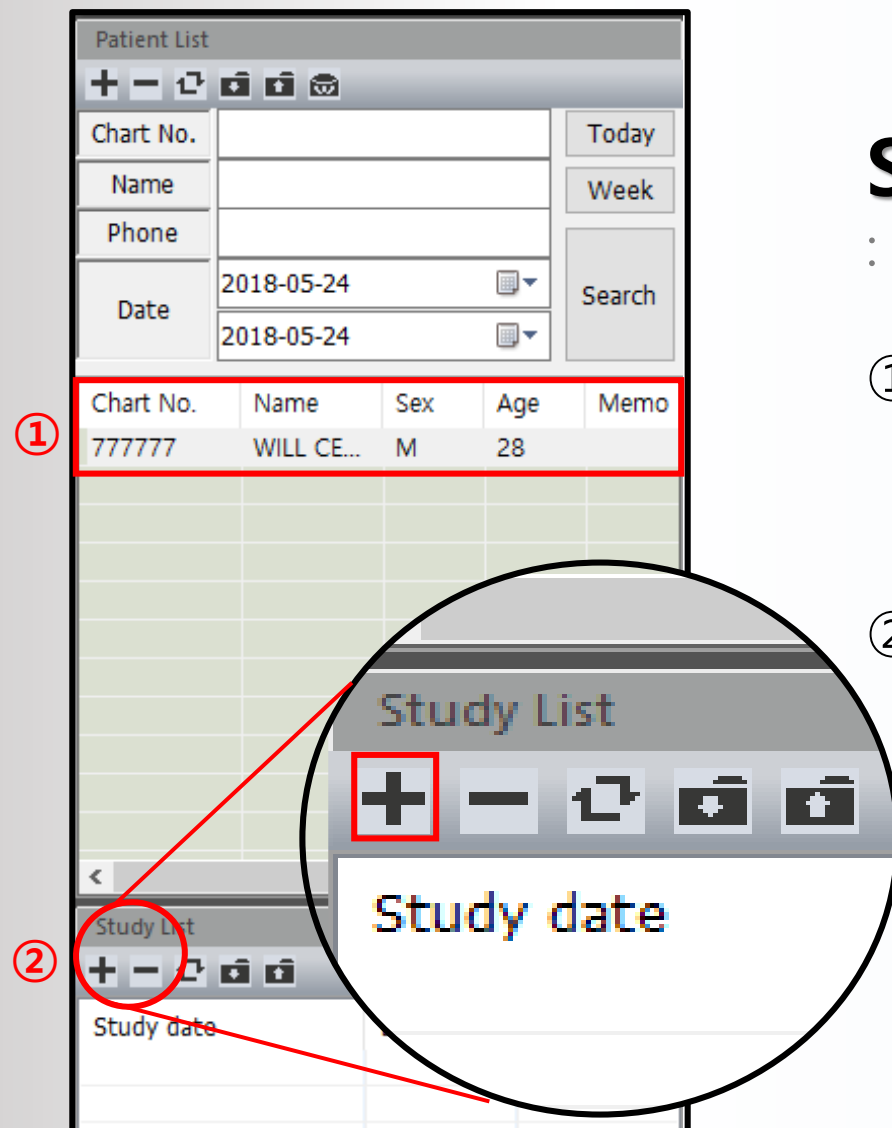
2) Manually import images

Step 2

: Add study for the patient

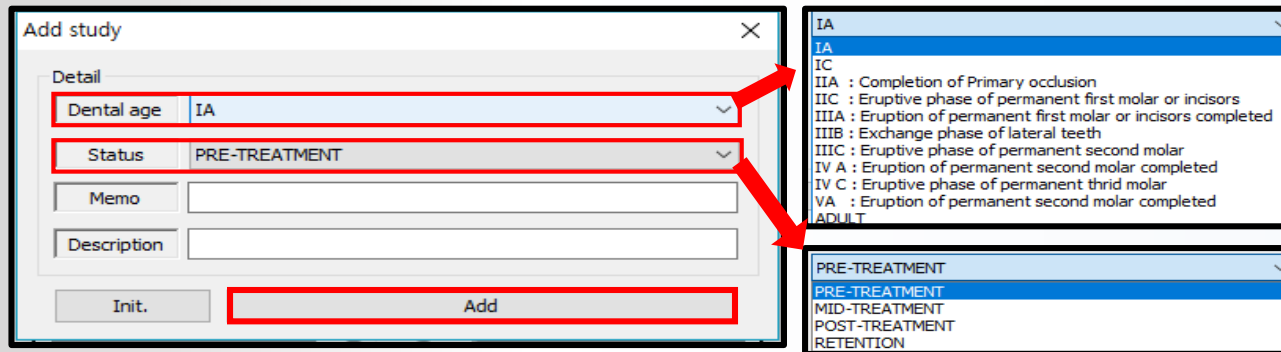
① Search and select the patient until you see the name highlighted in blue

② Click  plus symbol



2) Manually import images

③ Set Dental Age and Status



Add study

Detail

Dental age: IA

Status: PRE-TREATMENT

Memo:

Description:

Init. Add

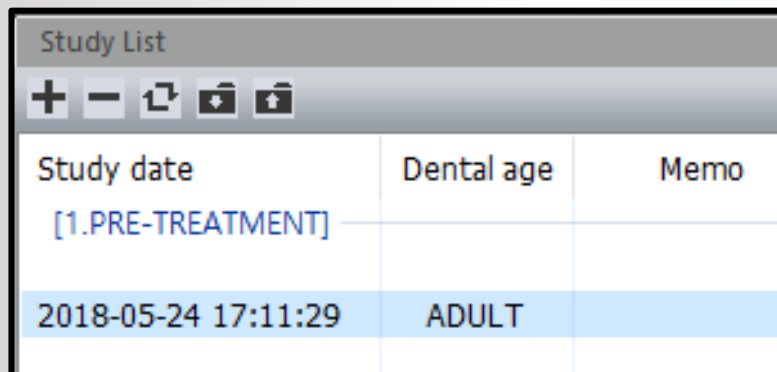
Dental Age List:

- IA
- IC
- IIA : Completion of Primary occlusion
- IIC : Eruptive phase of permanent first molar or incisors
- IIIA : Eruption of permanent first molar or incisors completed
- IIIB : Exchange phase of lateral teeth
- IIIC : Eruptive phase of permanent second molar
- IV A : Eruption of permanent second molar completed
- IV C : Eruptive phase of permanent third molar
- VA : Eruption of permanent second molar completed
- ADULT

Status List:

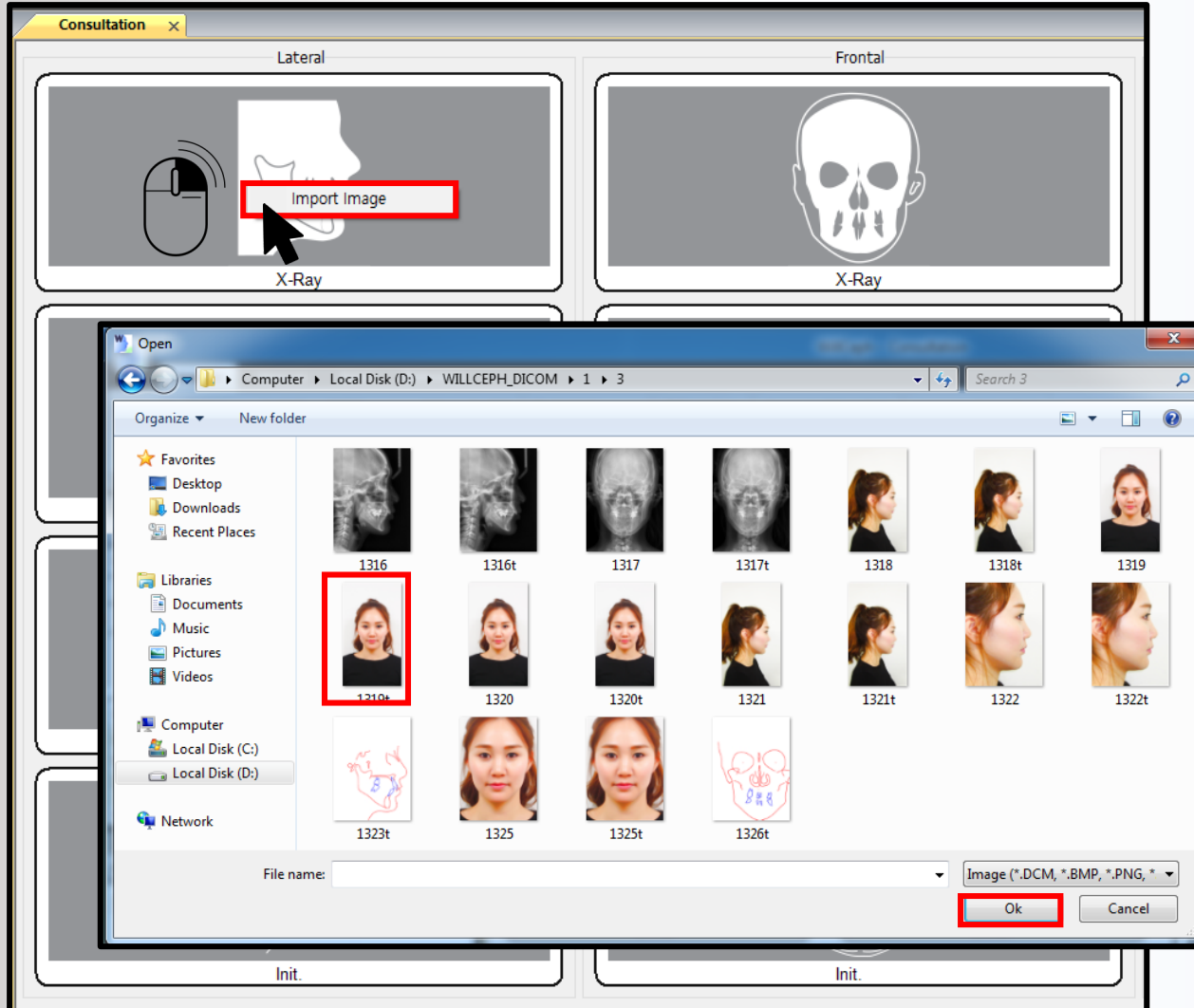
- PRE-TREATMENT
- MID-TREATMENT
- POST-TREATMENT
- RETENTION

④ Patient Registration Complete



Study date	Dental age	Memo
[1.PRE-TREATMENT]		
2018-05-24 17:11:29	ADULT	

2) Manually import images



Step 3

: Import images from Computer

- ① Right click  on a spot
- ② Click on "import image" to choose image from folder

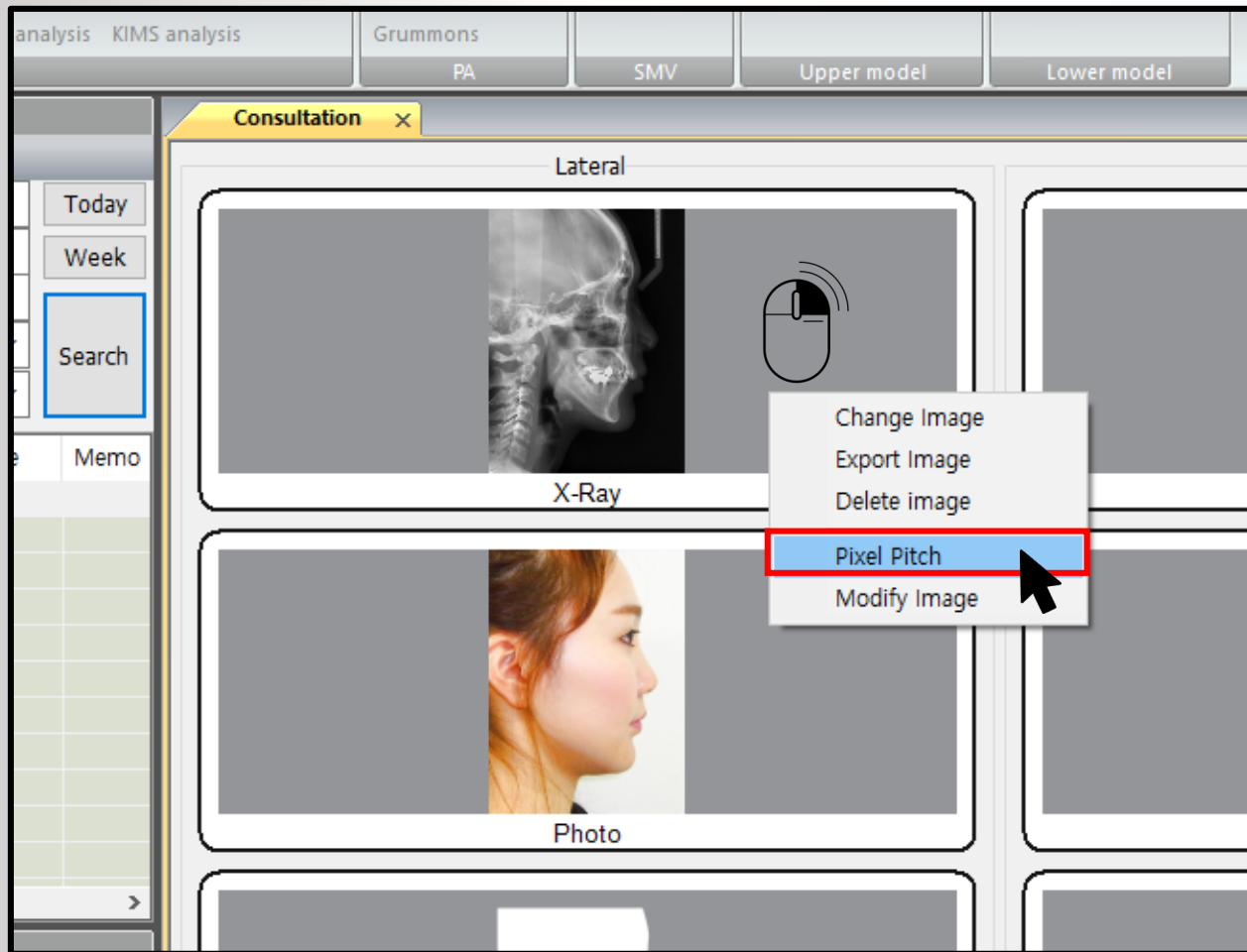
2) Manually import images

Step 4

: Manually Calibrate images for analysis

When images are imported from PC folder, Calibration must be performed

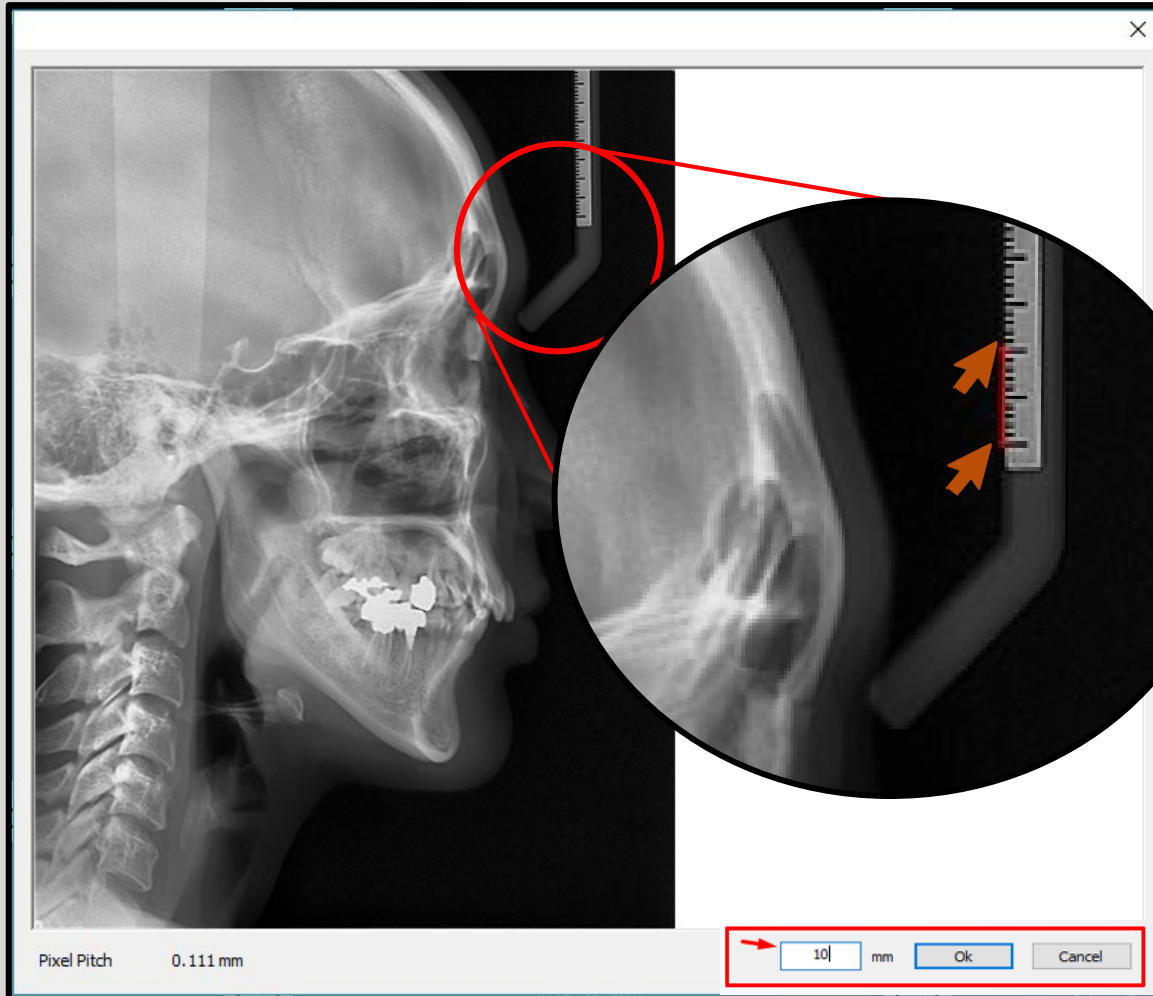
- ① After importing images, right click  on the x-ray image and select [Pixel Pitch]



2) Manually import images

Step 4

: Manually calibrate images for analysis



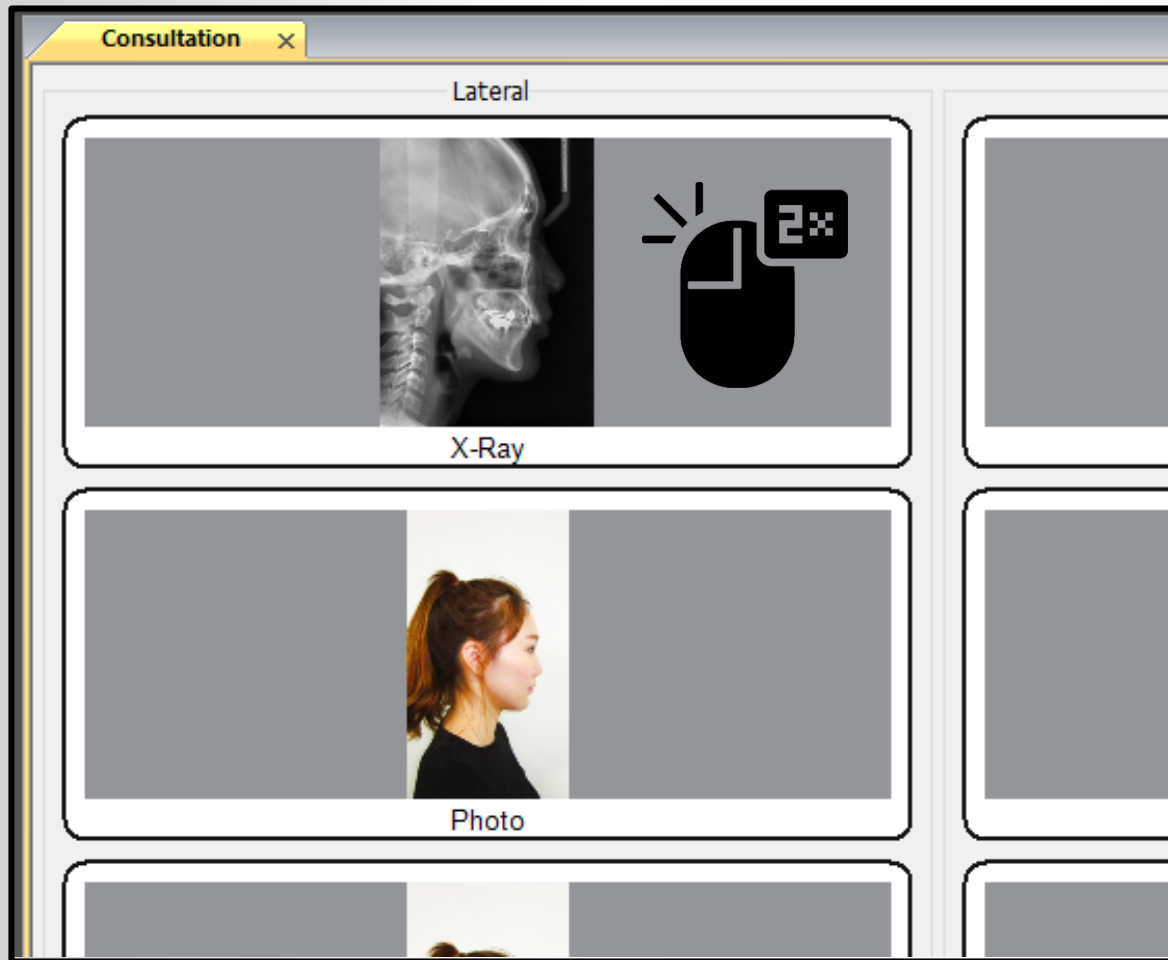
- ② Click on two locations on the ruler in the image and enter the distance between them below
- ③ Keep in mind the ruler is measured in millimeters. The image to the left illustrates the two points.
- ④ Once you identify both points you will manually enter in numerical value below.
- ⑤ Click Ok

2. Cephalometric Tracing

- 1) Trace Task Window Layout
- 2) Tracing Steps
- 3) Patient Photo Alignment



1) Trace Task Window Layout



To open Trace Task Window,
Double click on patient x-ray image

1) Trace Task Window Layout

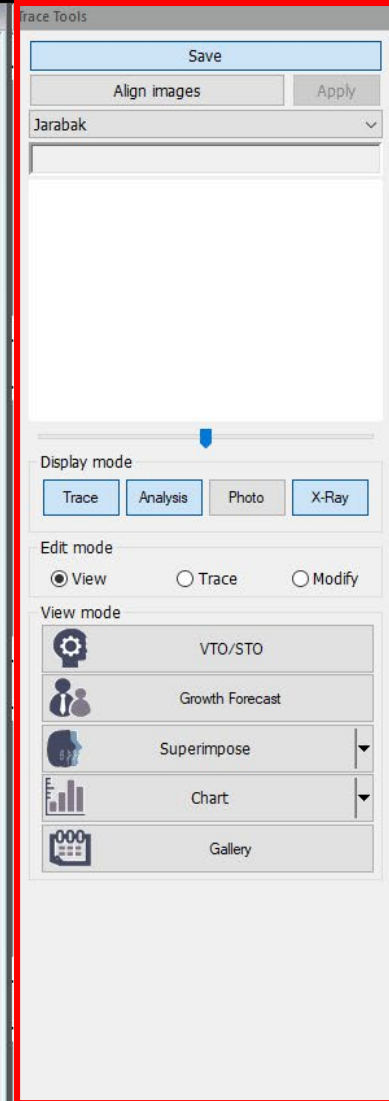
① Output Screen

This Window displays X-Ray, Photo or Tracing Image and is used to pick/delete/edit Landmark Points while tracing.



② Tool bar

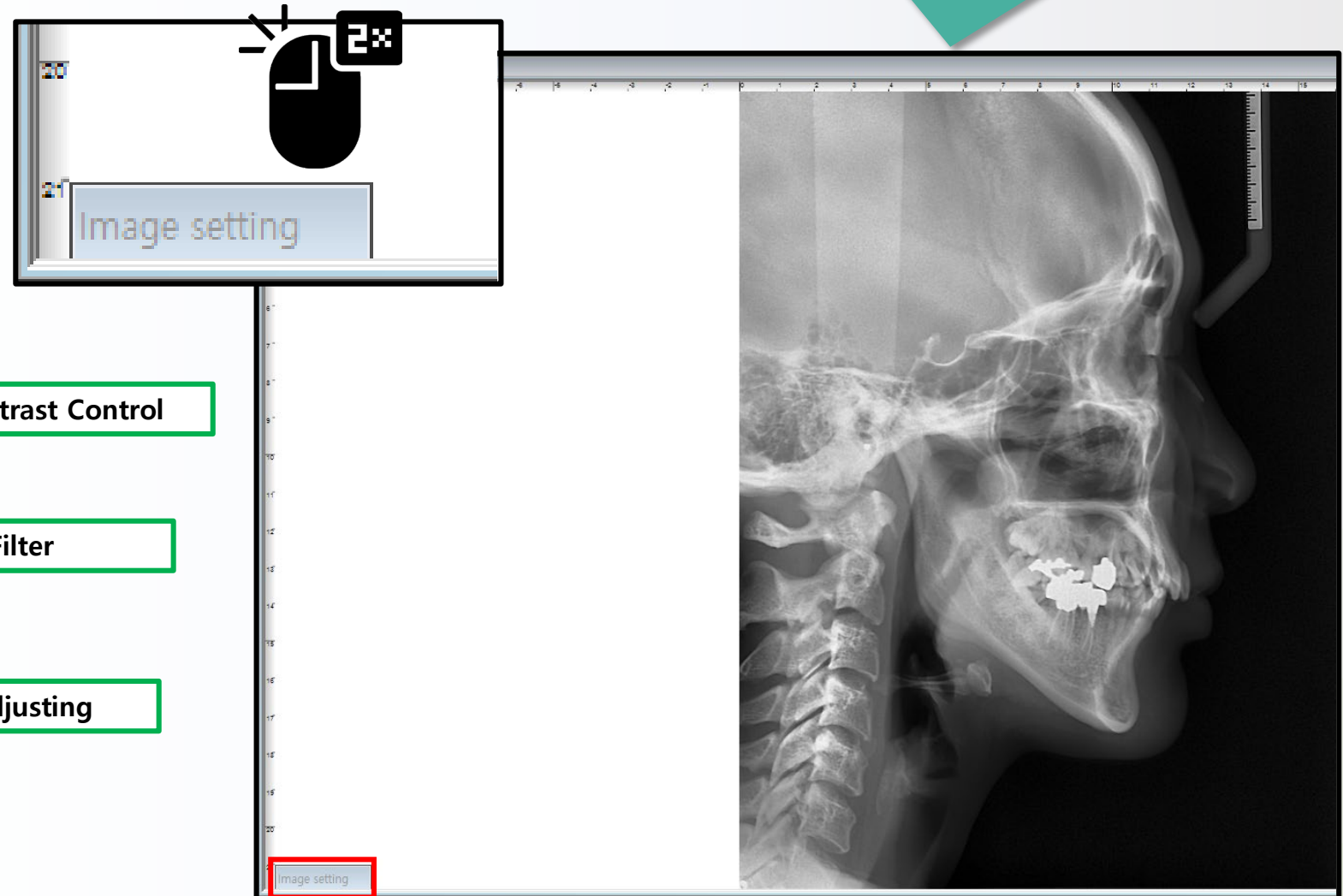
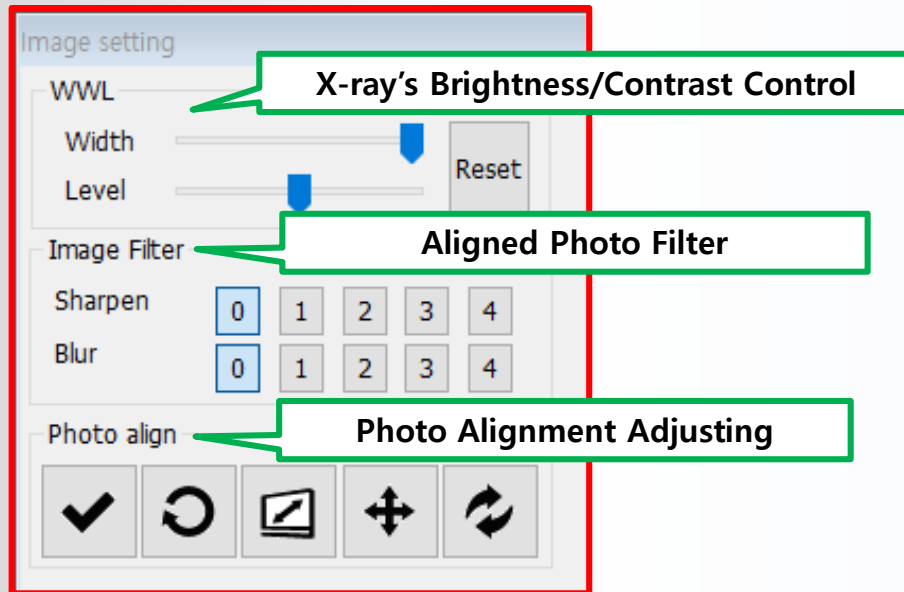
Tool bar contains tracing and analysis tools



1) Trace Task Window Layout

③ Image setting window

- Bottom left corner Double click to open image setting window

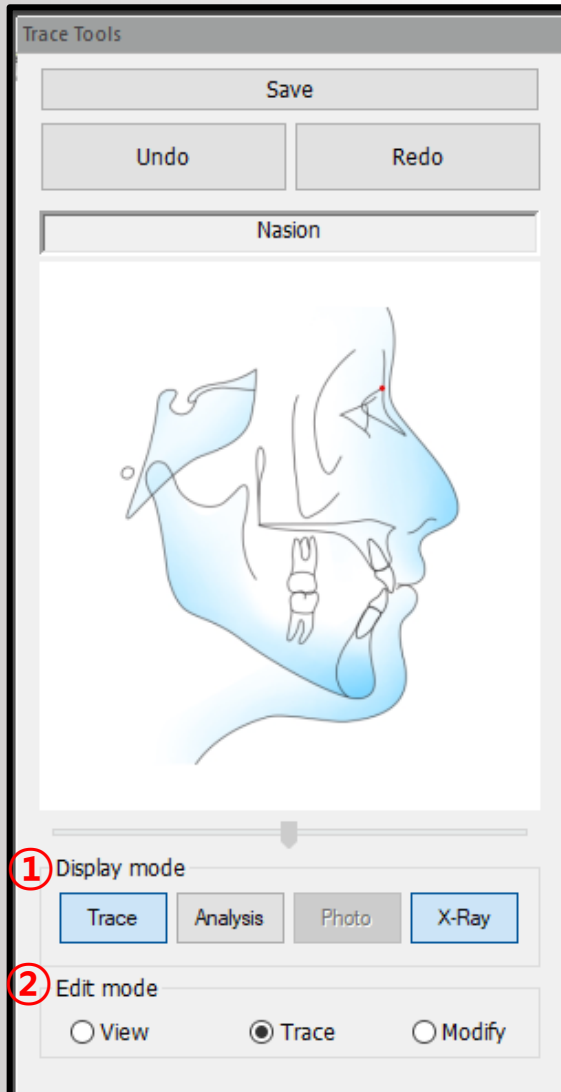


2) Tracing Steps

Step 1

: Trace Tools

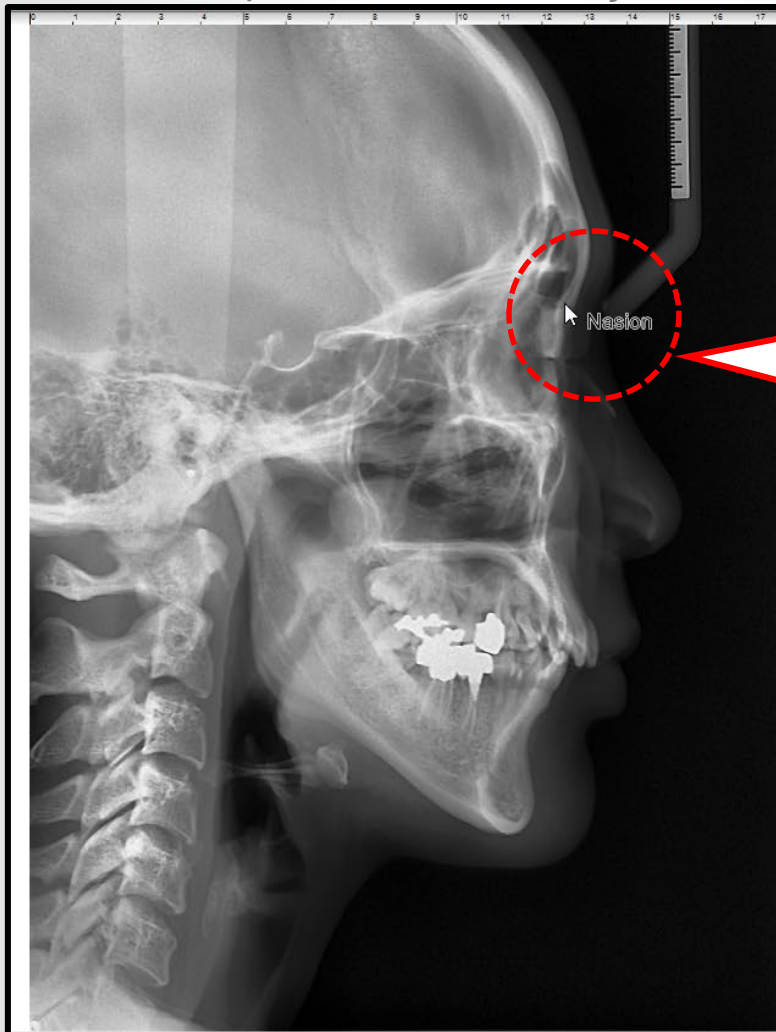
- ① Display mode
 - select [X-Ray], [Trace] for tracing (Deselect all other)
- ② Edit mode
 - select [Trace] for tracing



2) Tracing Steps

Step 2

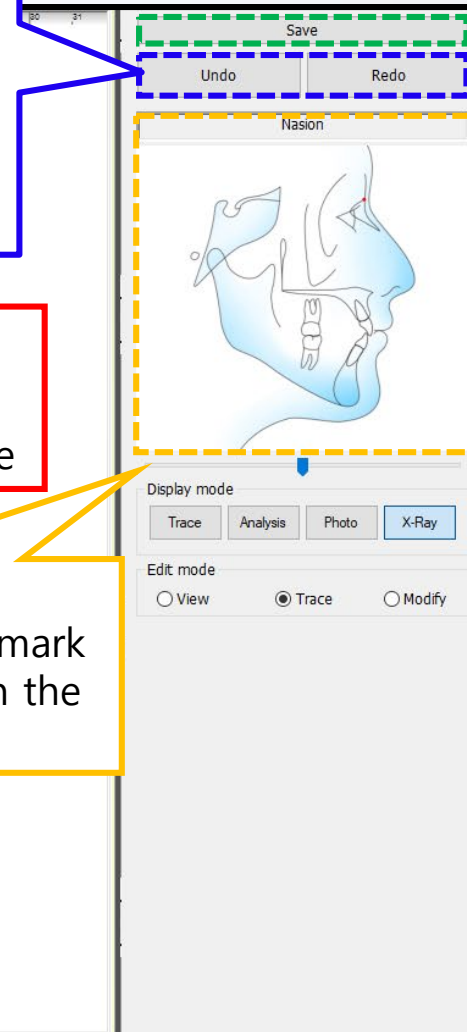
: Trace on the Cephalometric X-ray



(3)
Click "Undo" to go back
one step
Click "Redo" to go one
step forward

(2)
Click on the landmark
following the guide image

(1)
The location of the landmark
to be picked is shown in the
guide window



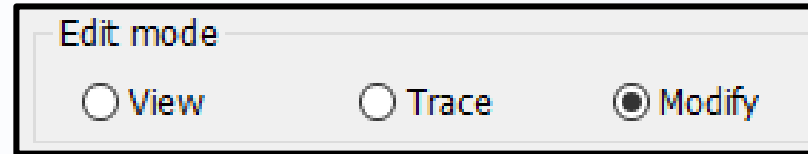
(4)
Click [Save]
to save the final result

2) Tracing Steps

Step 3

: Modify Tracing line

① After tracing, Select [Modify] in [Edit mode]



② To modify Point/Outline

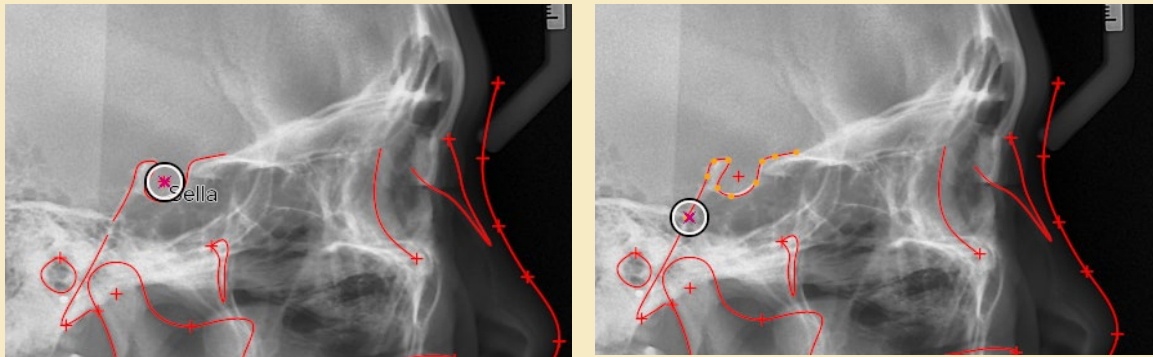
A. Landmark/Outline point

- Hover the mouse cursor over until (x) appears
- Click and drag (x) to the desired location

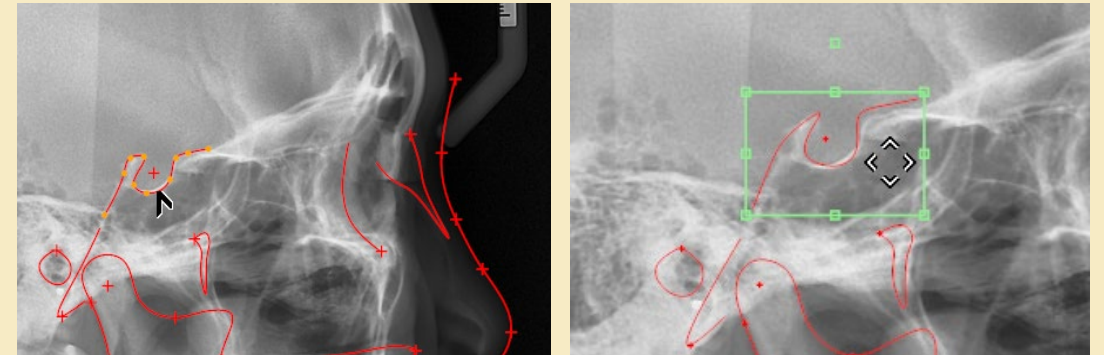
B. Entire Outline

- Click on the outline
- Drag the selection box to the desired location

A. Point



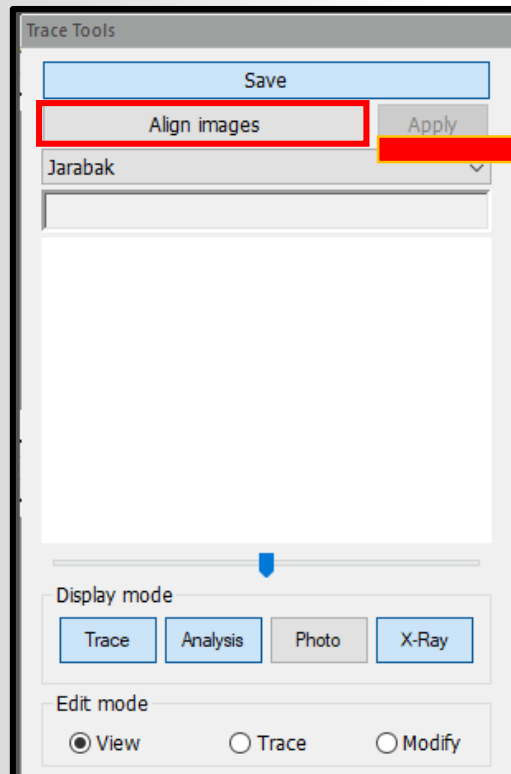
B. Outline



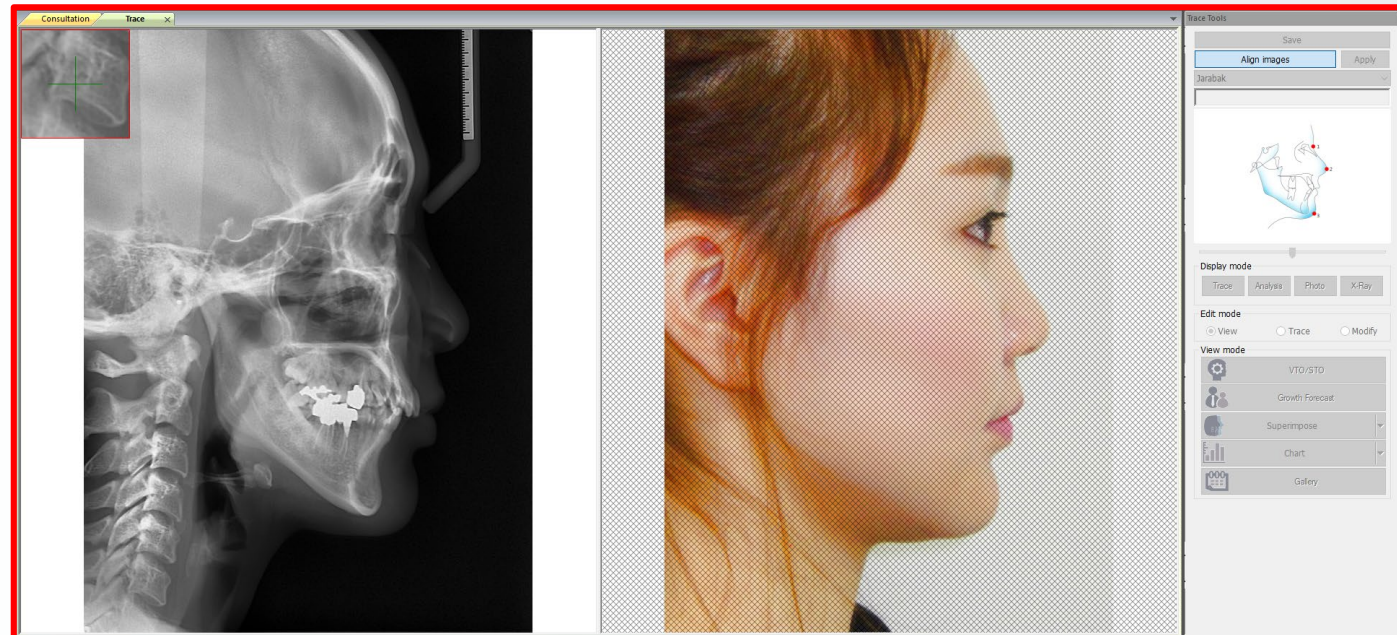
3) Patient Photo Alignment

Photo Alignment

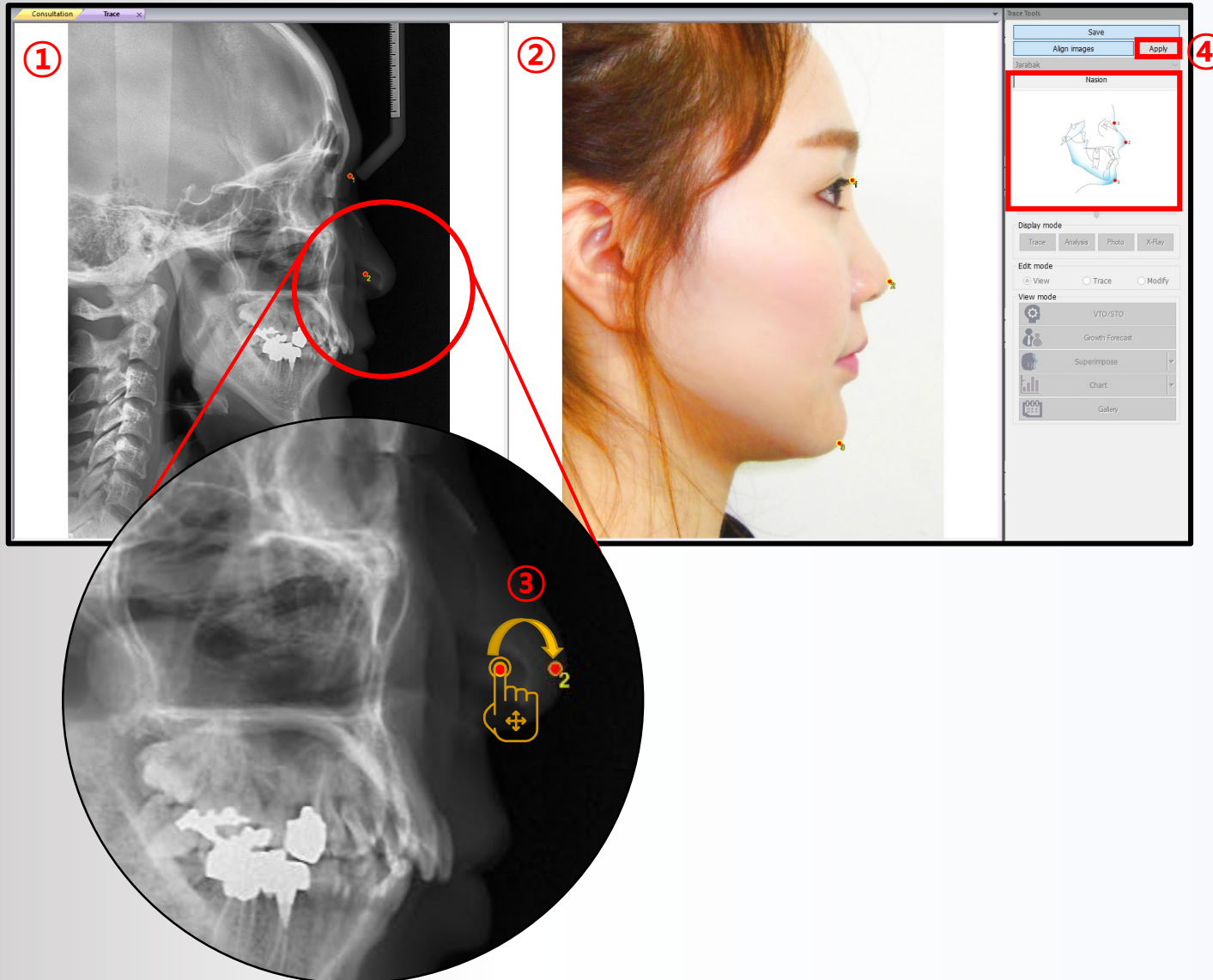
To use VTO/STO (Visualized Treatment Objective/Surgical Treatment Objective) feature, align photo with X-ray



Click **Align images** the mode changes to image align mode



3) Patient Photo Alignment



Step 1

: Rough Photo Alignment

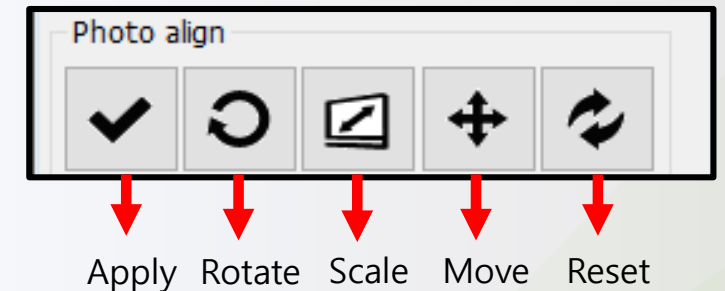
- ① Pick 3 dots on the X-ray image to match the 3 dots in the guide window
- ② Mark the same 3 dots on the patient photo
- ③ If necessary, click and drag to correct dot position
- ④ Click  to finish alignment

3) Patient Photo Alignment

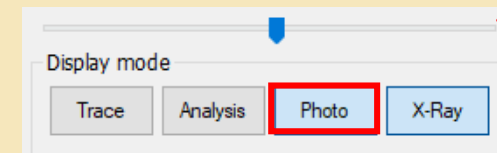
Step 2

: Adjusting Photo Alignment

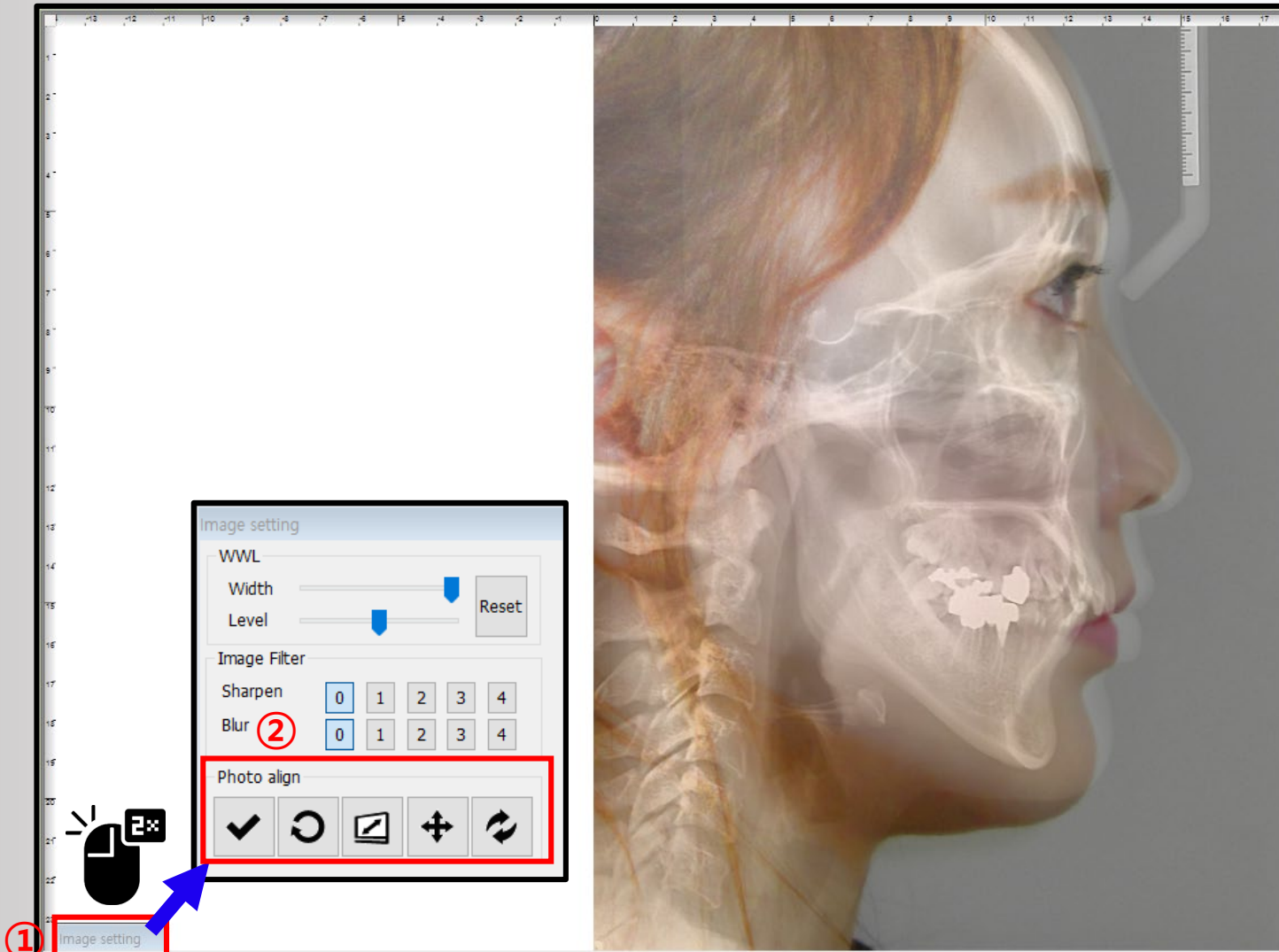
- ① Double click "image setting" to open image setting window
- ② Adjust aligned photo using **Photo align** tools



Select the Photo button in [Display mode] to display the photo layer

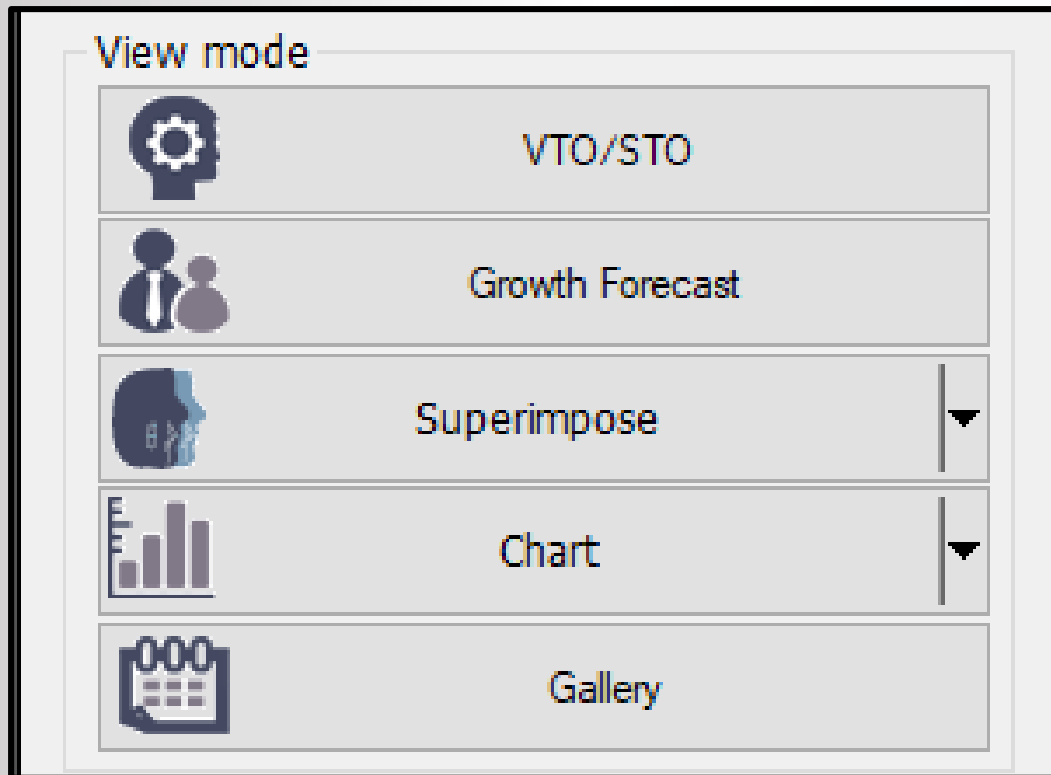


Superimposed photo layer transparency control



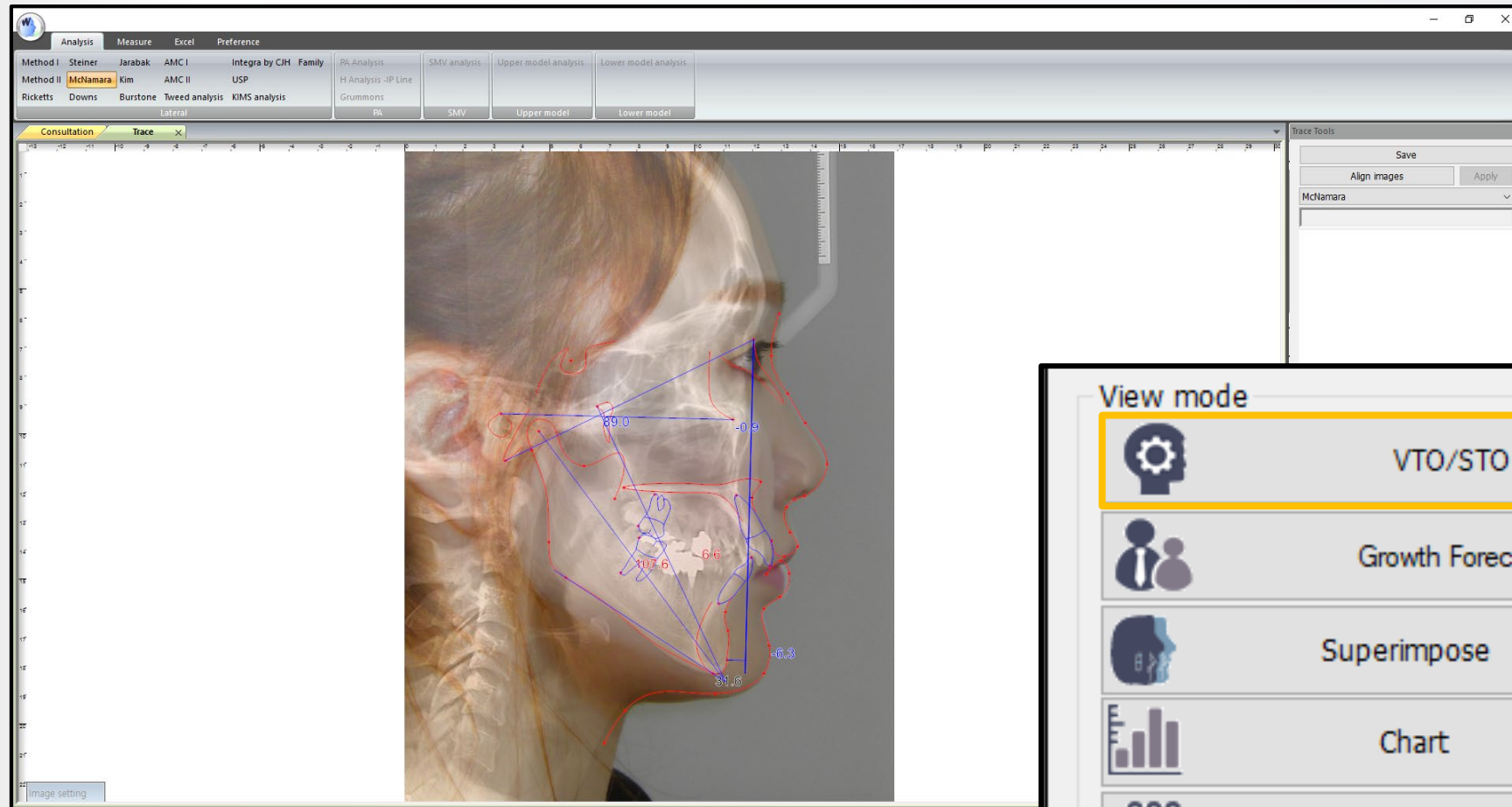
Cephalometric Analysis

View Mode



- **VTO/STO** : Facial contour projection, before and after treatment
- **Growth Forecast** : Growth prediction for up to 5 years
- **Superimposition** : Comparison of the trace line of the before and after treatment
- **Chart** : Analysis calculation
- **Gallery** : Summary of the images from each function in the various layouts

1) VTO/STO : Predicted Simulation Before & After



View mode



VTO/STO



Growth Forecast



Superimpose



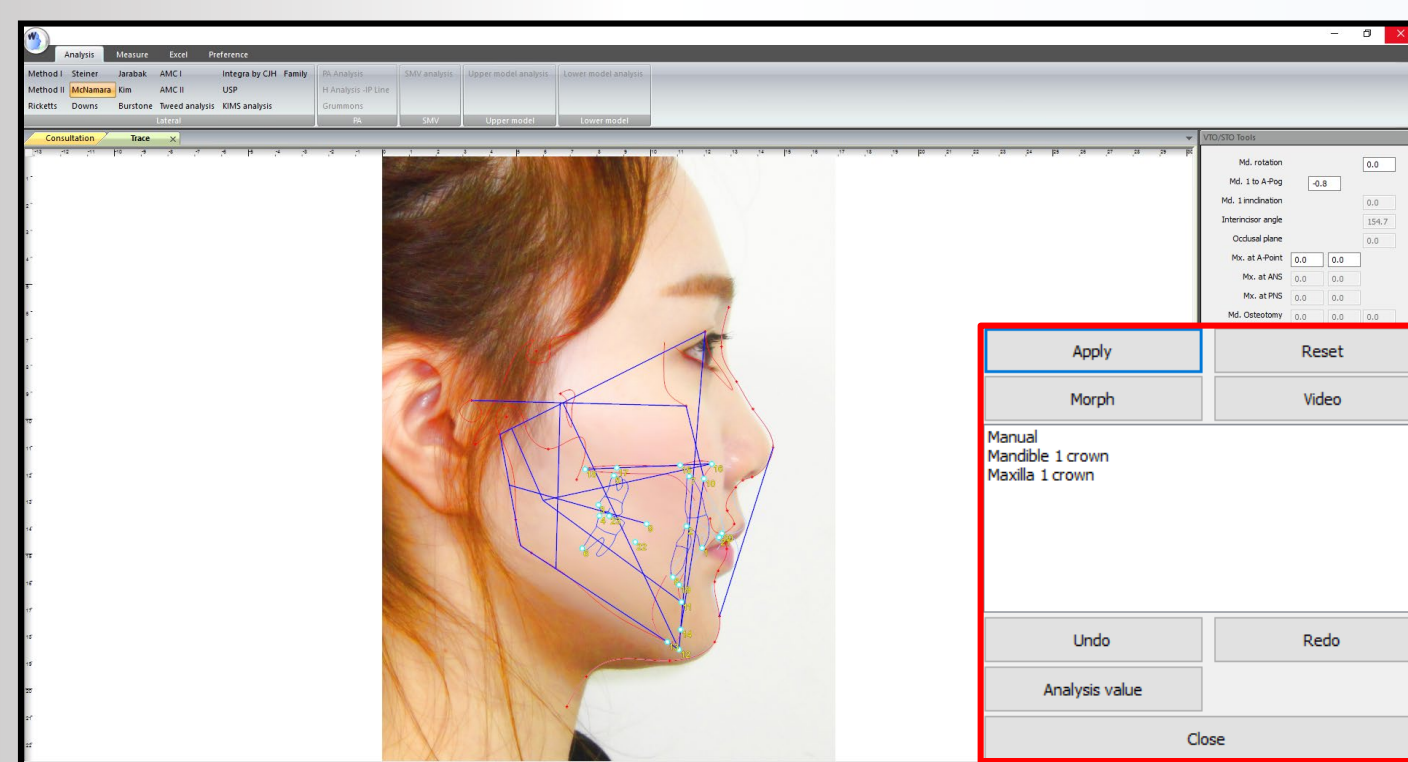
Chart



Gallery

1) VTO/STO : Predicted Simulation Before & After

Tools

**Apply**

: Applying changed value

Reset

: Resetting the value

Morph

: Performing after treatment simulation

Video

: Viewing simulation video clip and comparison image.

Undo**Redo**

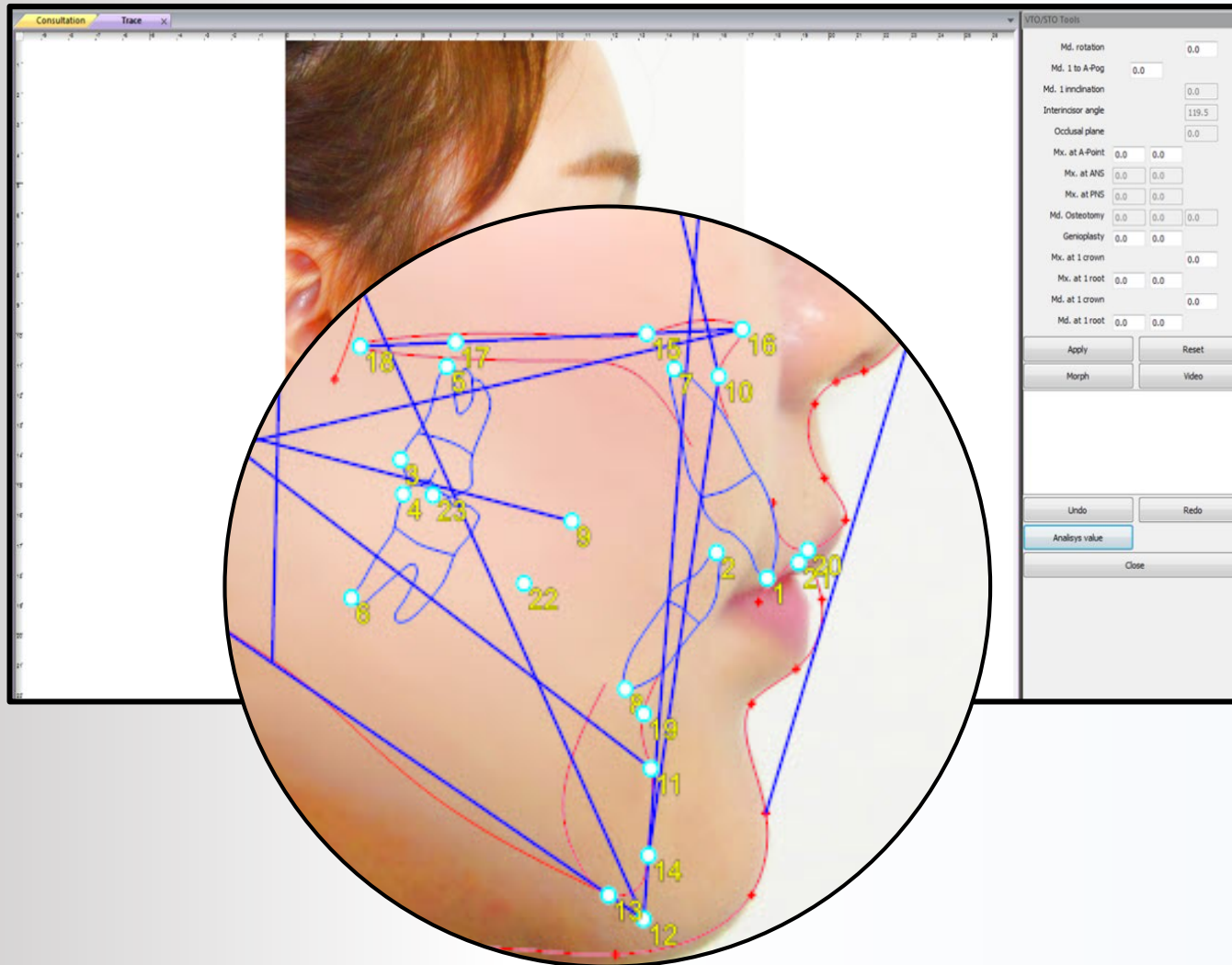
: Undo and Redo a step

Analysis value

: Displaying the analysis value

1) VTO/STO : Predicted Simulation Before & After

Controls

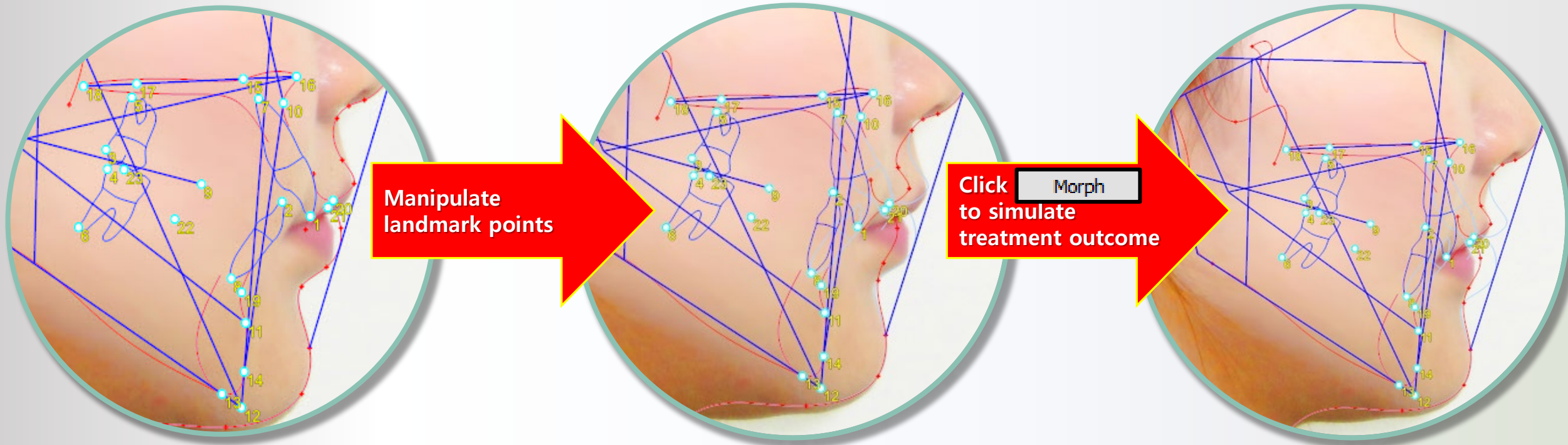


- Hover the cursor over each landmark point to enable control
- Depending on the landmark, one of the following functions will become available
- Each number in the table corresponds to a specific landmarks

Function	Landmarks
 Move	5(Maxilla 6 Root), 6(Mandible 6 Root), 7(Maxilla 1 Root), 8(Mandible 1 Root), 10(A-Point), 11(Protuberance Menti), 19(B-Point)
 Rotate	1(Maxilla 1 Crown), 2(Mandible 1 Crown), 3(Maxilla 6 Distal), 4(Mandible 6 Distal), 12, 6(Anterior Nasal Point), 18(Posterior Nasal Spine), 20(stms), 21(stmi)
 Cut	13(Menton), 14(Pogonion), 15, 17, 22, 23

1) VTO/STO : Predicted Simulation Before & After

Procedure



Before treatment

- Original tracing line
- Original facial contour

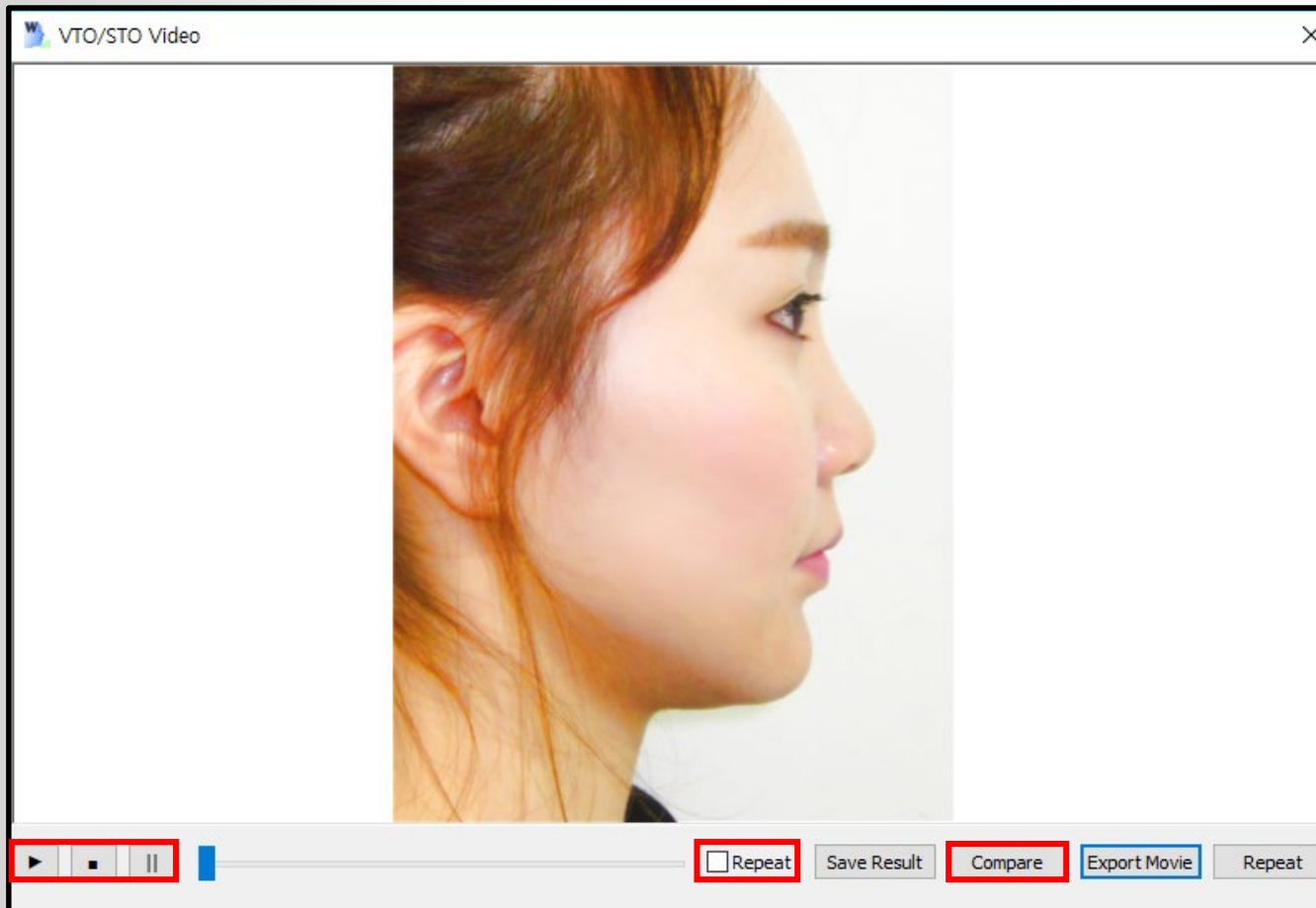
- Adjusted tracing line
- Original facial contour

Prediction of After treatment

- Adjusted tracing line
- Adjusted facial contour

1) VTO/STO : Predicted Simulation Before & After

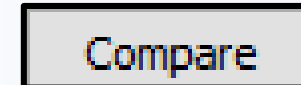
3. Simulation Video



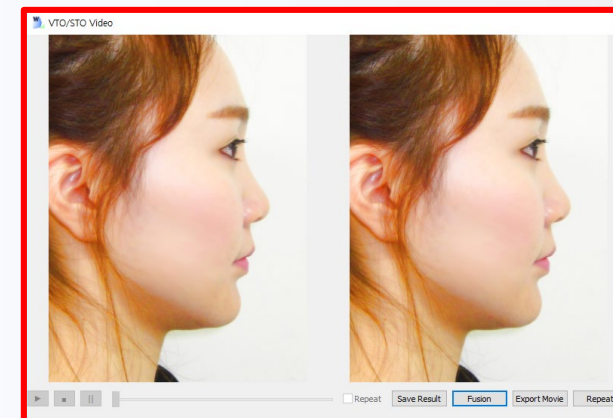
: Play, Stop , Pause the simulation video



: Repeat the simulation video



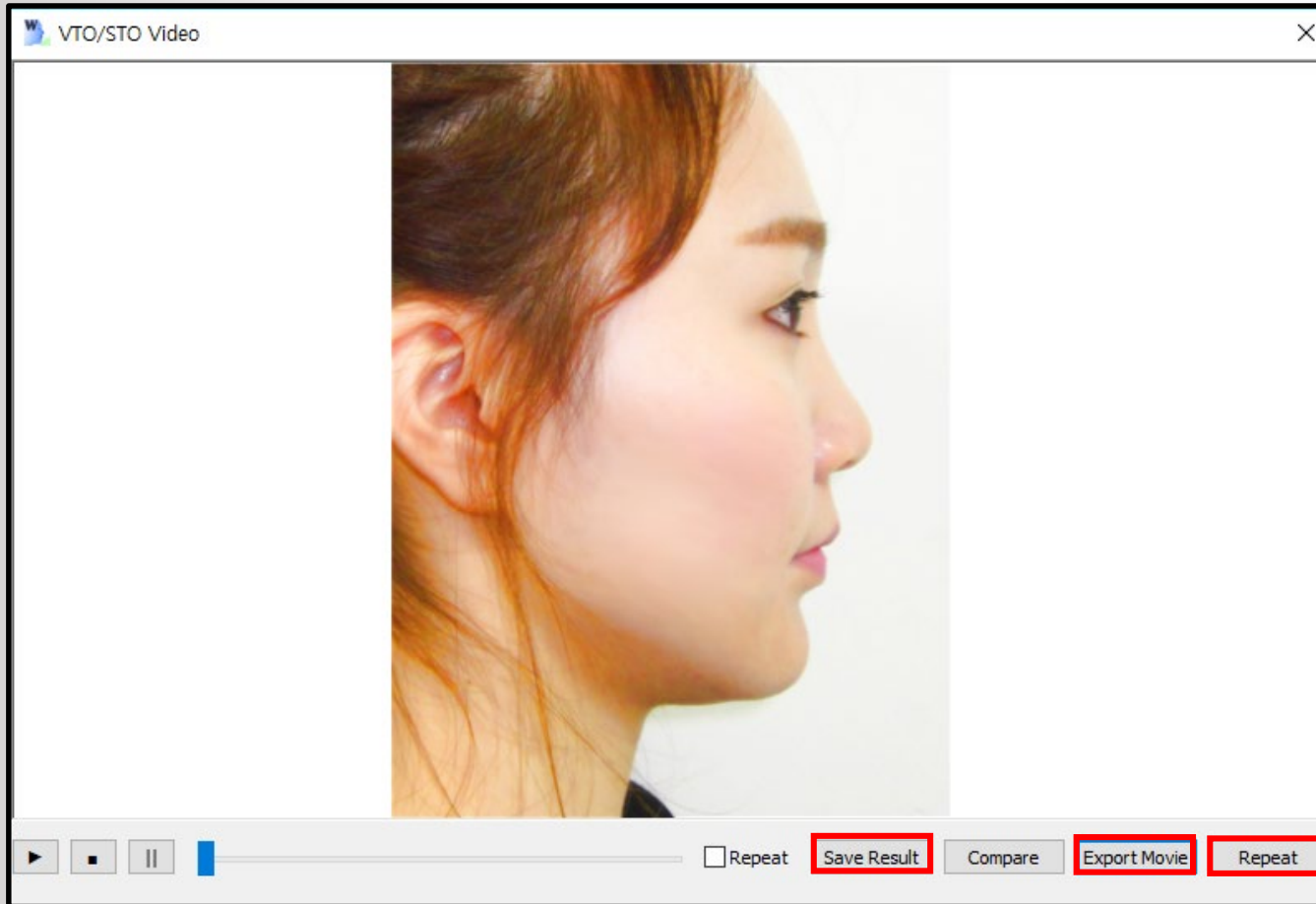
: View before / after treatment



* Click [Fusion] to go back

1) VTO/STO : Predicted Simulation Before & After

3. Simulation Video

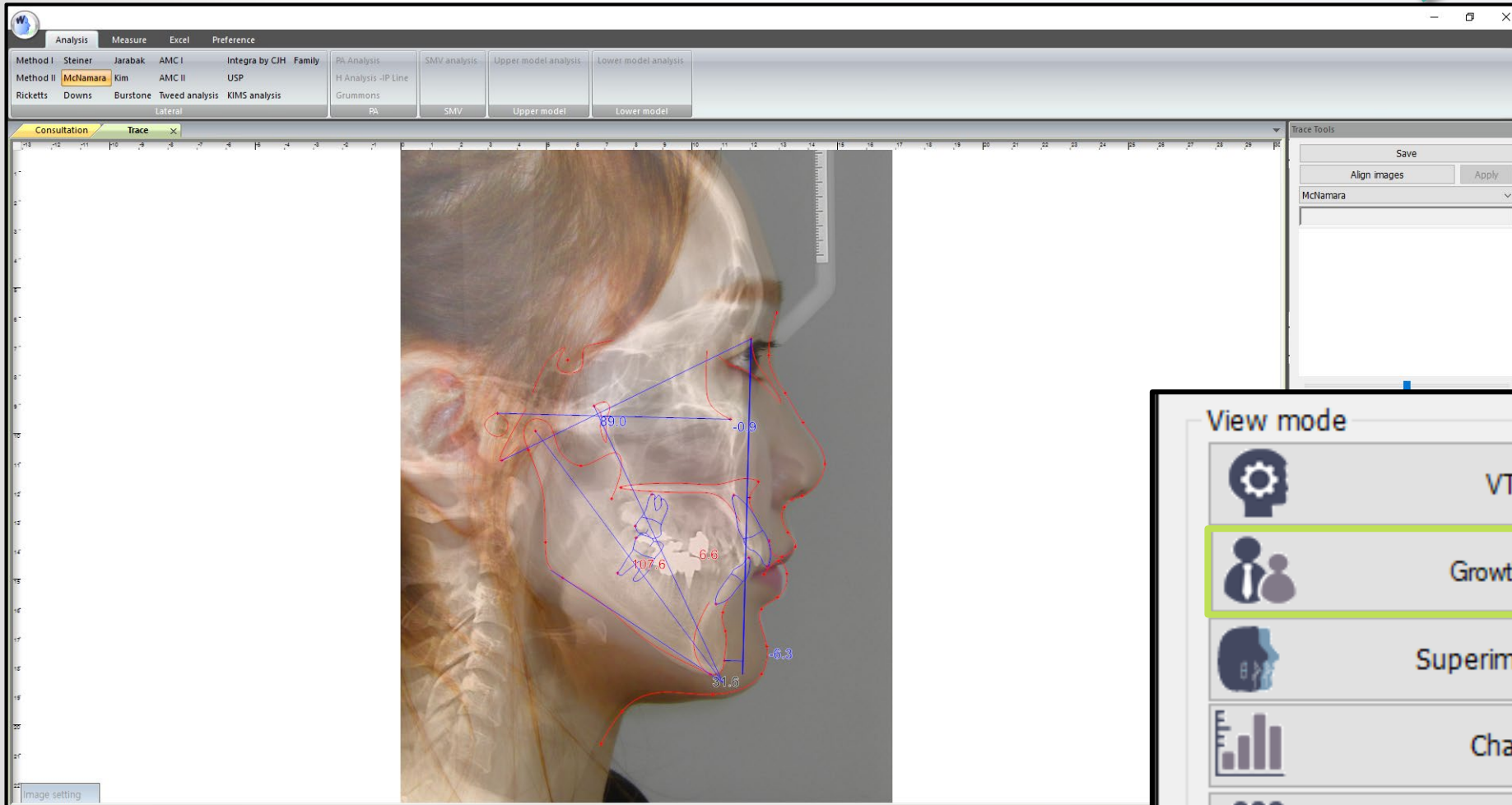


Save Result : Save the before and after Comparison image at ETC category of [Consultation] tab

Export Movie : Save simulation video as AVI file

Repeat : Repeat the procedure from the beginning. Same meaning as close the window.

2) Growth Forecast : Prediction based on analysis



View mode



VTO/STO



Growth Forecast



Superimpose

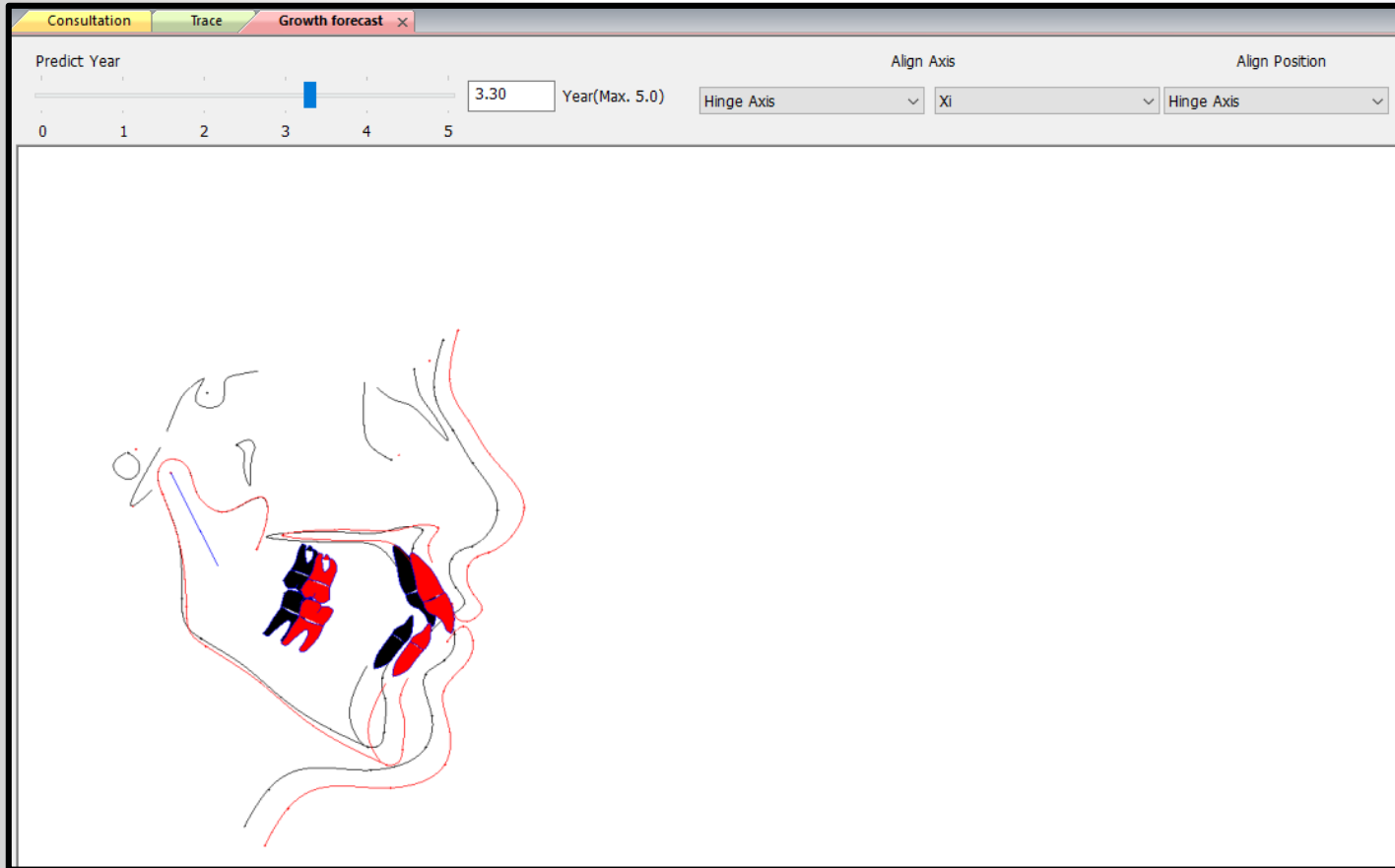


Chart



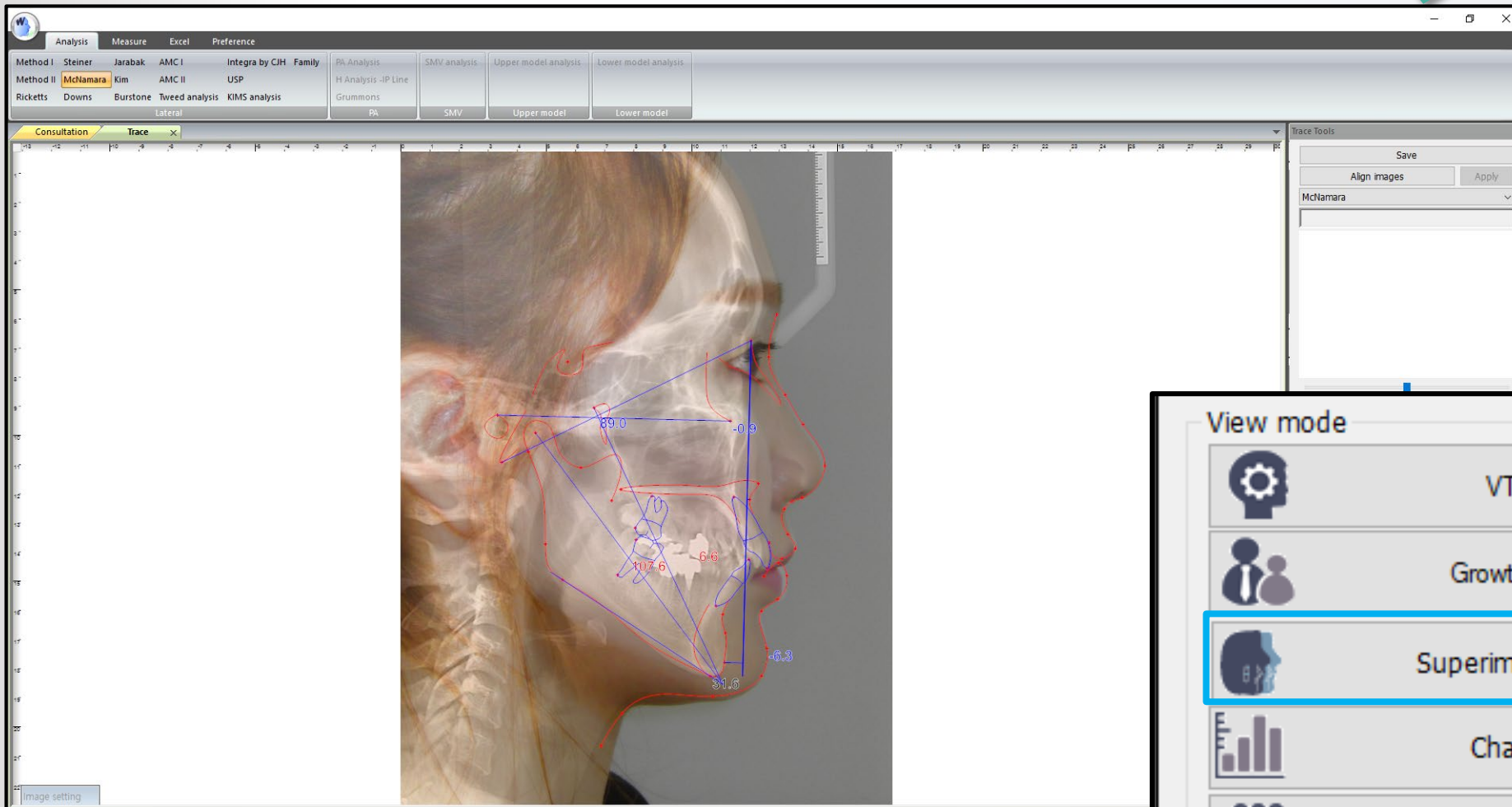
Gallery

2) Growth Forecast

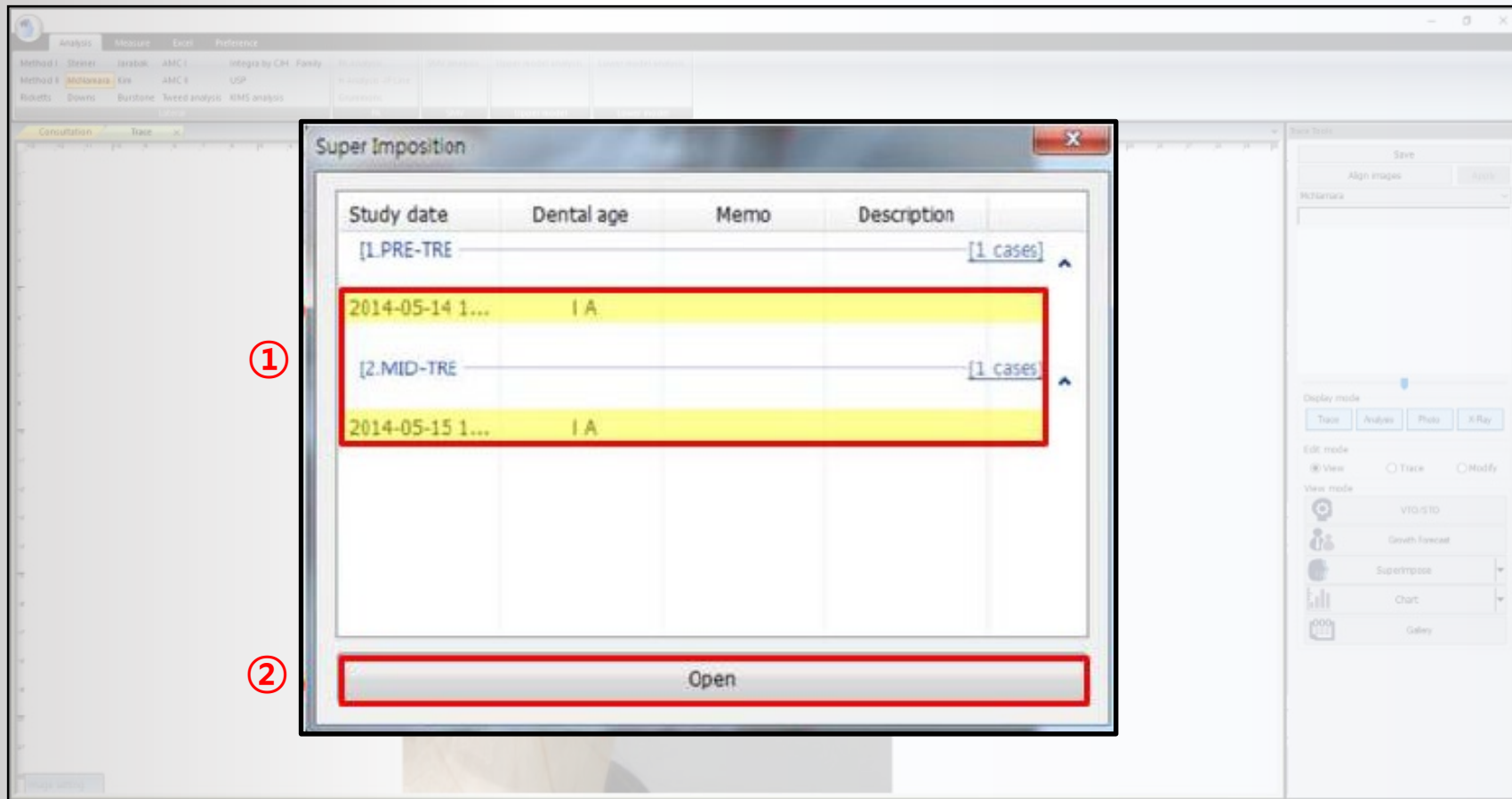


- ① Set up [Align Axis] and [Align Position]
- ② Predict : Use the scroll bar to predict the amount.
- ③ Amount allowed is up to 5years

3) Superimpose : Overlay for Comparison



3) Superimpose : Overlay for Comparison

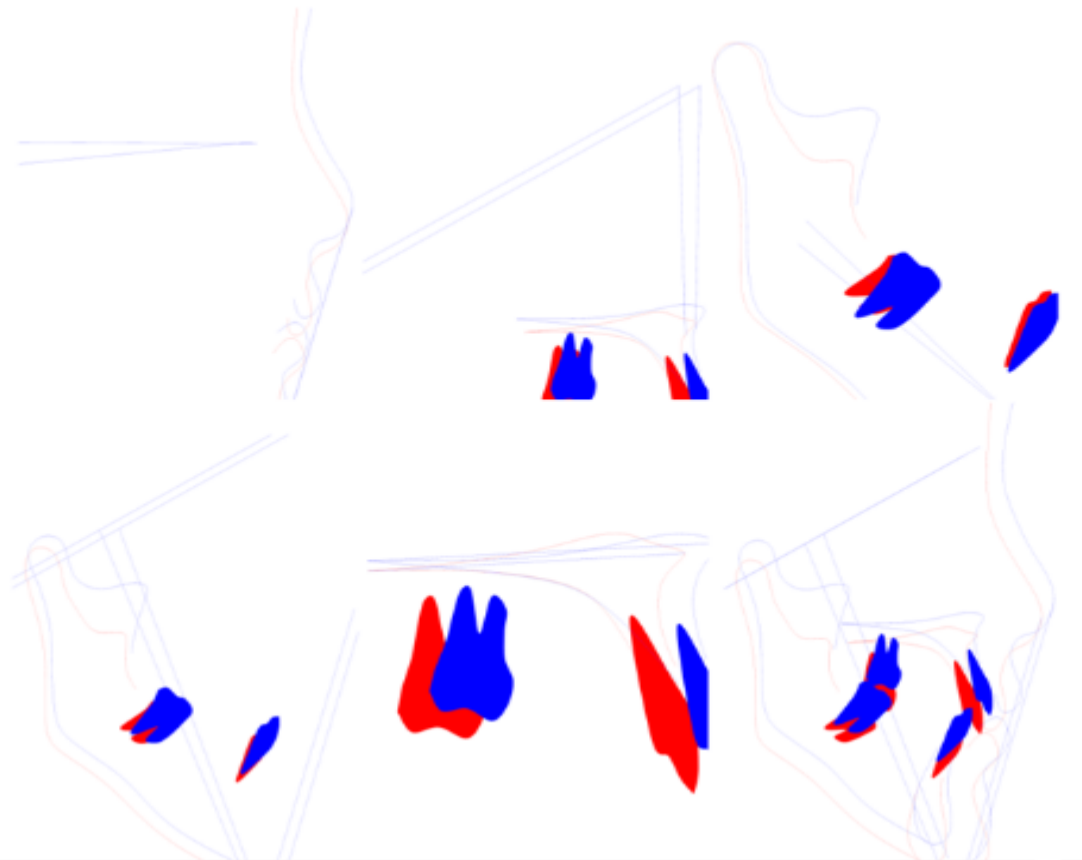


- ① Select multiple studies while holding **ctrl** key
- ② Click [Open]

3) Superimpose : Overlay for Comparison

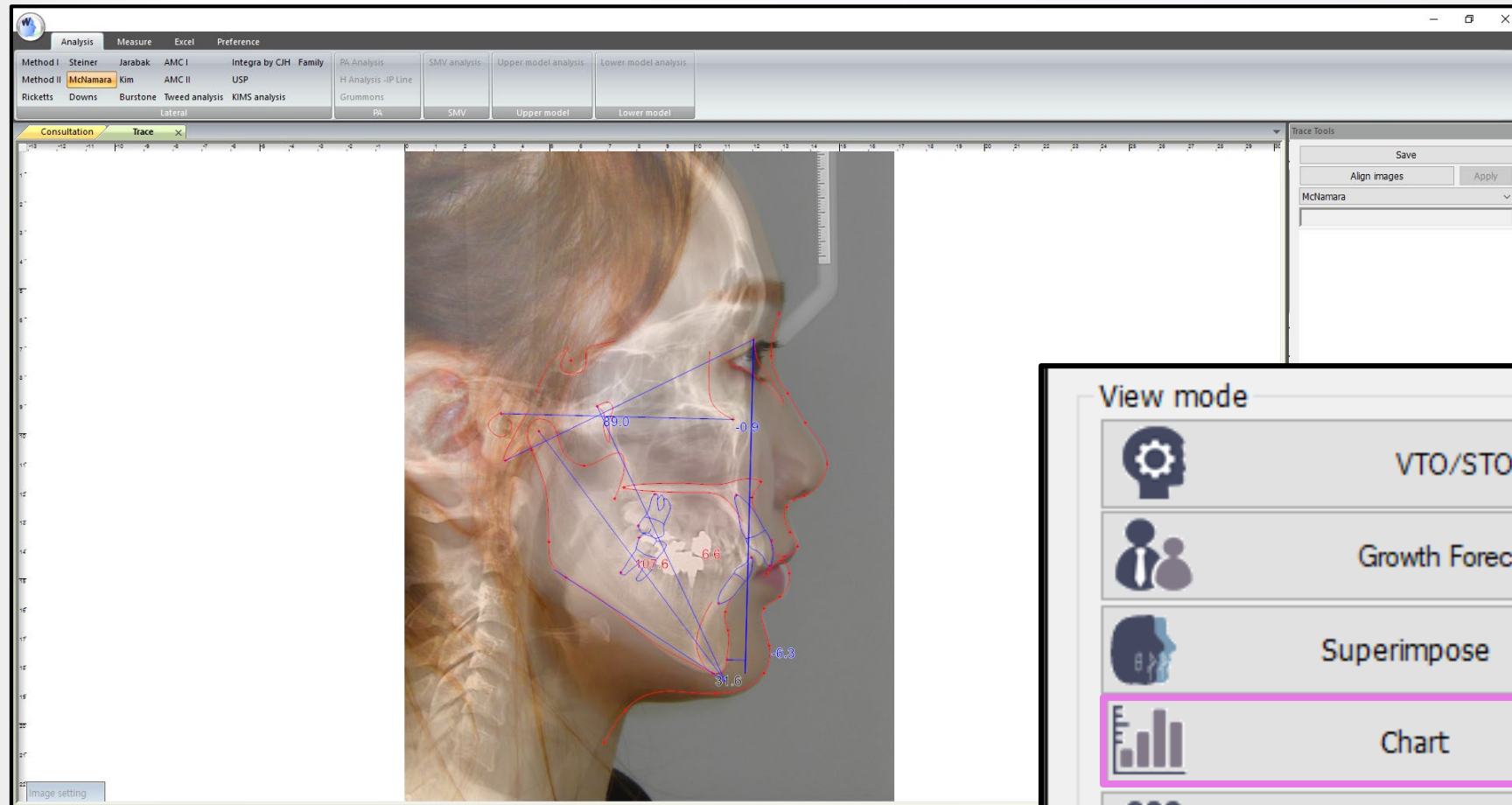
2014-05-15 13:21:13

2014-05-14 17:38:07



- ③ Result : Trace date color and image color match
- ④ You can turn each slide by clicking on the page

4) Chart : Analysis Calculation



View mode



VTO/STO



Growth Forecast



Superimpose

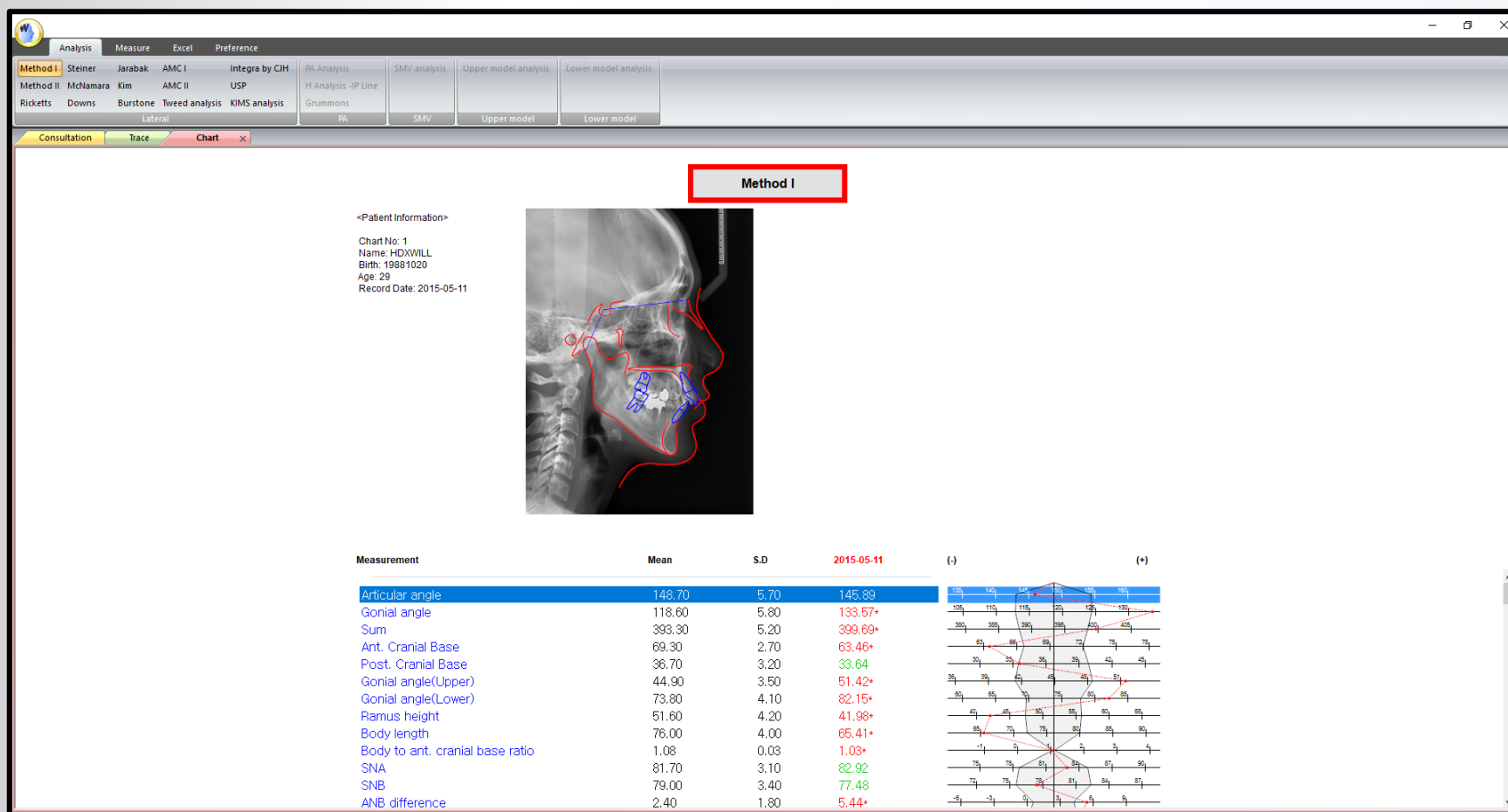


Chart



Gallery

4) Chart : Analysis Calculation



① Click **Method I** and select an analysis method of preference

Select method

Method I

Method II

Ricketts

Steiner

McNamara

Downs

Jarabak

Kim

Burstone

AMC I

AMC II

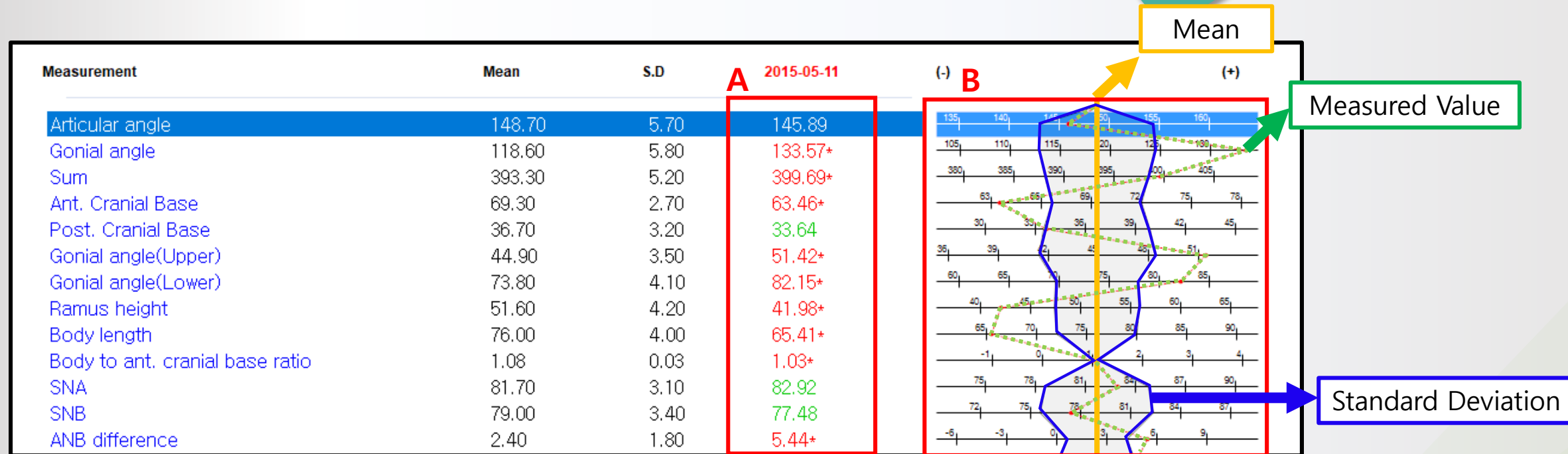
Tweed analysis

Integra by CJH

USP

KIMS analysis

4) Chart : Analysis Calculation



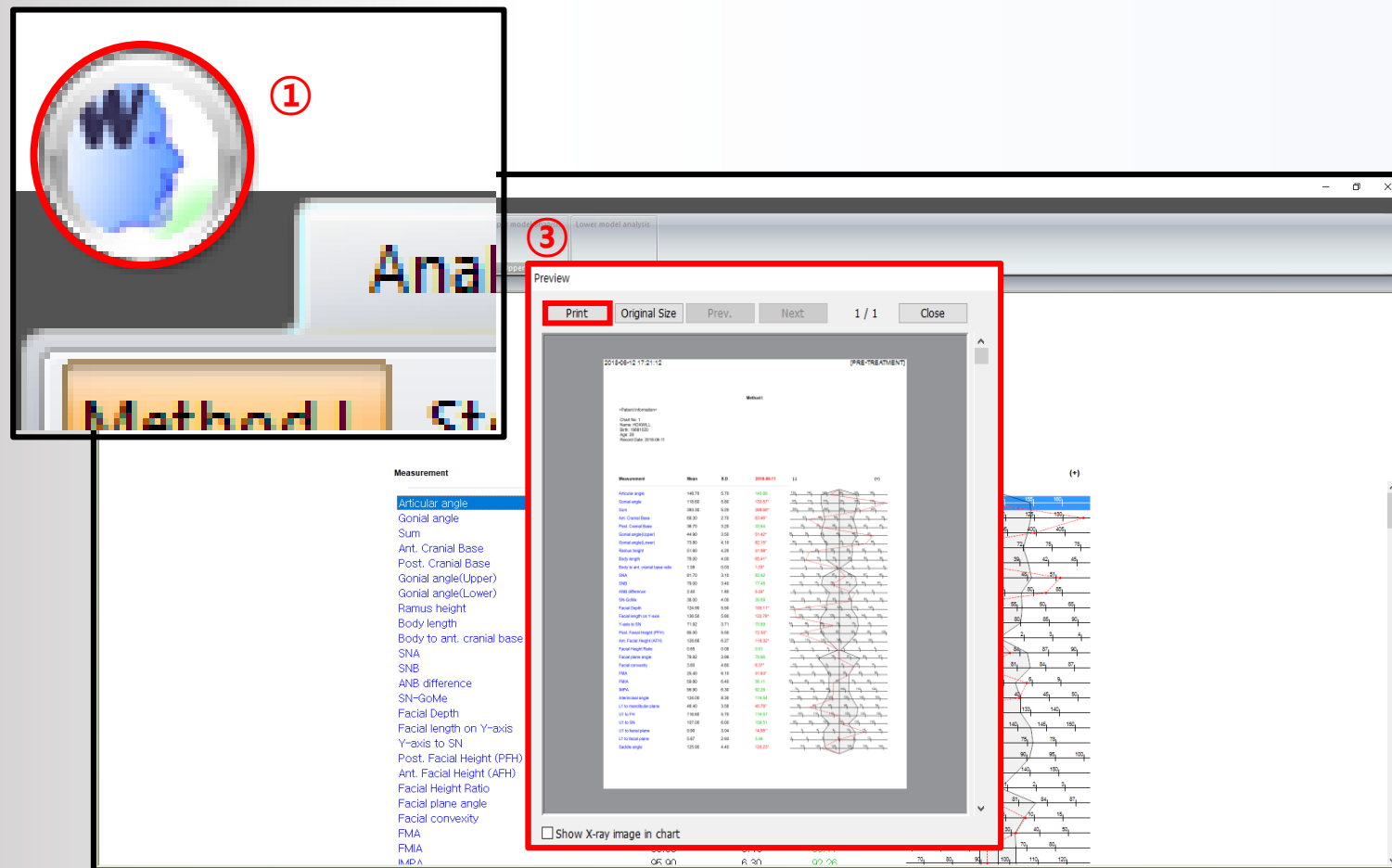
A. Measured Value

- Green Value : The patient's measured value is within the standard deviation of the means
- Red Value : The patient's measured value is beyond the range of the standard deviation

B. Graph

- Represents the means, standard deviations, and the measured values of each specific measurement item

4) Chart : Analysis Calculation



Measurement

- Articular angle
- Gonial angle
- Sum
- Ant. Cranial Base
- Post. Cranial Base
- Gonial angle(Upper)
- Gonial angle(Lower)
- Ramus height
- Body length
- Body to ant. cranial base
- SNA
- SNB
- ANB difference
- SN-GoMe
- Facial Depth
- Facial length on Y-axis
- Y-axis to SN
- Post. Facial Height (PFH)
- Ant. Facial Height (AFH)
- Facial Height Ratio
- Facial plane angle
- Facial convexity
- FMA
- FMA
- IMPA

Preview

Print Original Size Prev. Next 1 / 1 Close

2014-08-12 17:21:12 (PHE-TREATMENT)

Measurement

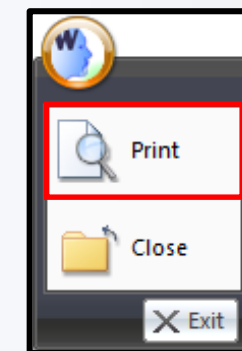
Measurement	Unit	Value	SD	2014-08-12	LI	PI
Articular angle	145.73	5.73	145.50			
Gonial angle	119.08	5.58	119.07			
Sum	264.81	5.25	264.57			
Ant. Cranial Base	18.30	2.75	18.47			
Post. Cranial Base	36.70	3.00	36.64			
Gonial angle(Upper)	44.80	3.58	44.47			
Gonial angle(Lower)	73.80	4.11	73.57			
Ramus height	51.80	4.28	51.87			
Body length	70.30	4.55	69.87			
Body to ant. cranial base	1.08	0.08	1.07			
SNA	81.70	2.11	82.02			
SNB	70.30	3.44	71.44			
ANB difference	11.40	1.66	10.57			
SN-GoMe	38.30	4.50	38.00			
Facial Depth	120.00	5.00	120.11			
Facial length on Y-axis	120.00	5.00	120.11			
Y-axis to SN	71.02	3.71	70.89			
Post. Facial Height (PFH)	36.70	3.00	36.64			
Ant. Facial Height (AFH)	18.30	2.75	18.47			
Facial Height Ratio	1.08	0.08	1.07			
Facial plane angle	70.30	3.44	71.44			
Facial convexity	1.08	0.08	1.07			
FMA	38.30	4.50	38.00			
IMPA	38.30	4.50	38.00			
Gonial angle	119.08	5.58	119.07			
Gonial angle(Upper)	44.80	3.58	44.47			
Gonial angle(Lower)	73.80	4.11	73.57			
Ramus height	51.80	4.28	51.87			
Body length	70.30	4.55	69.87			
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Facial Height Ratio	1.08	0.08	1.07			
Facial plane angle	70.30	3.44	71.44			
Facial convexity	1.08	0.08	1.07			
FMA	38.30	4.50	38.00			
IMPA	38.30	4.50	38.00			

☐ Show X-ray image in chart

2. Print Chart

① Click  at the right side of the top

② Click [Print]



③ Click [Print] of the [Preview] window

Cephalometric Analysis

5) Gallery

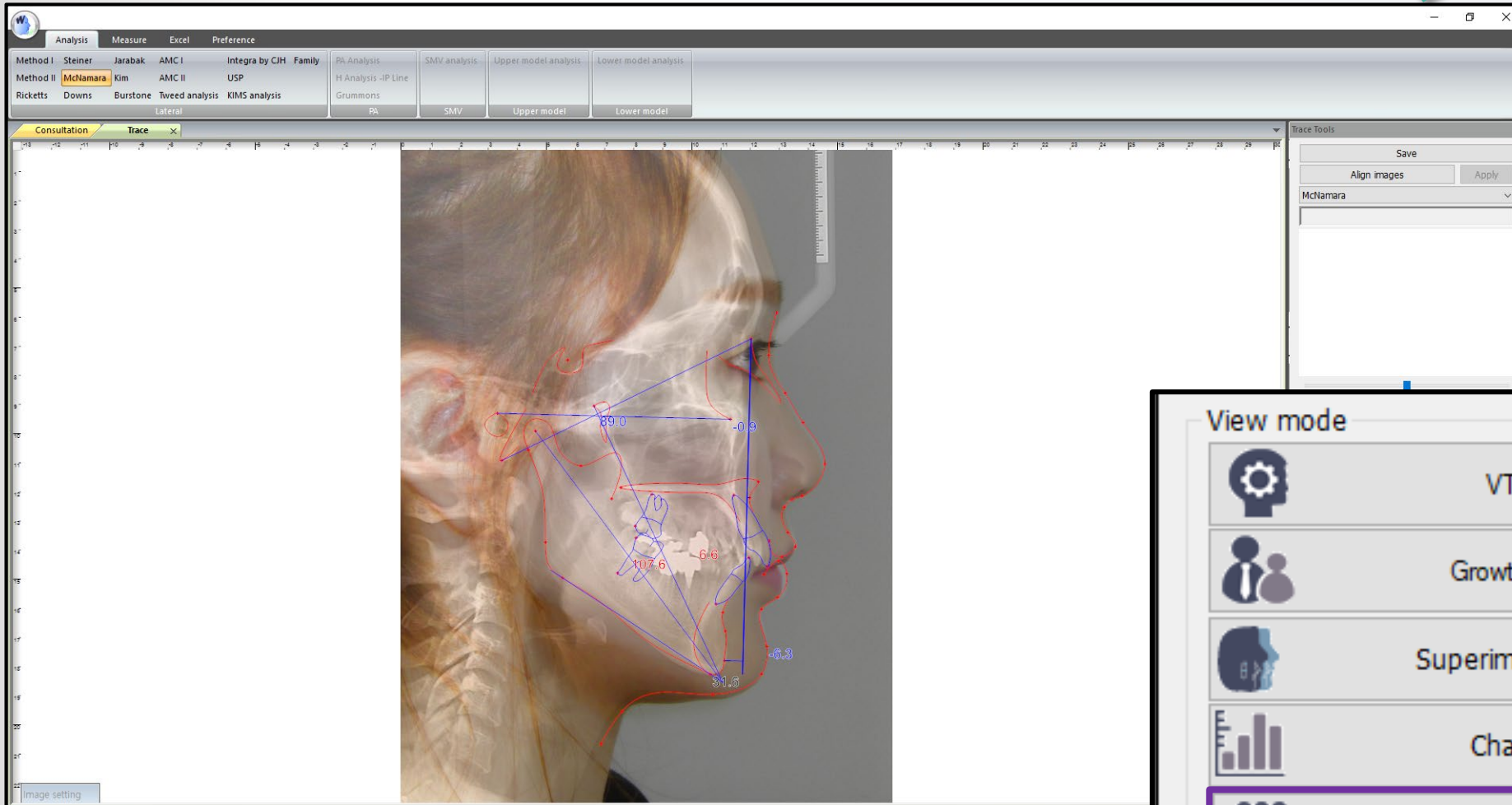
VTO/STO

Growth
Forecast

Super
imposition

Chart

Gallery



View mode



VTO/STO



Growth Forecast



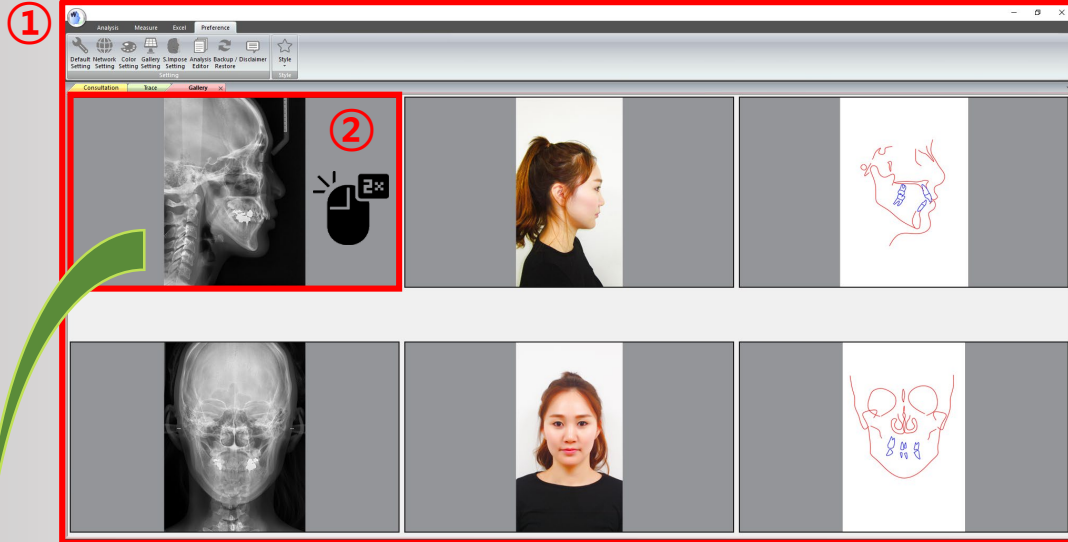
Superimpose



Chart



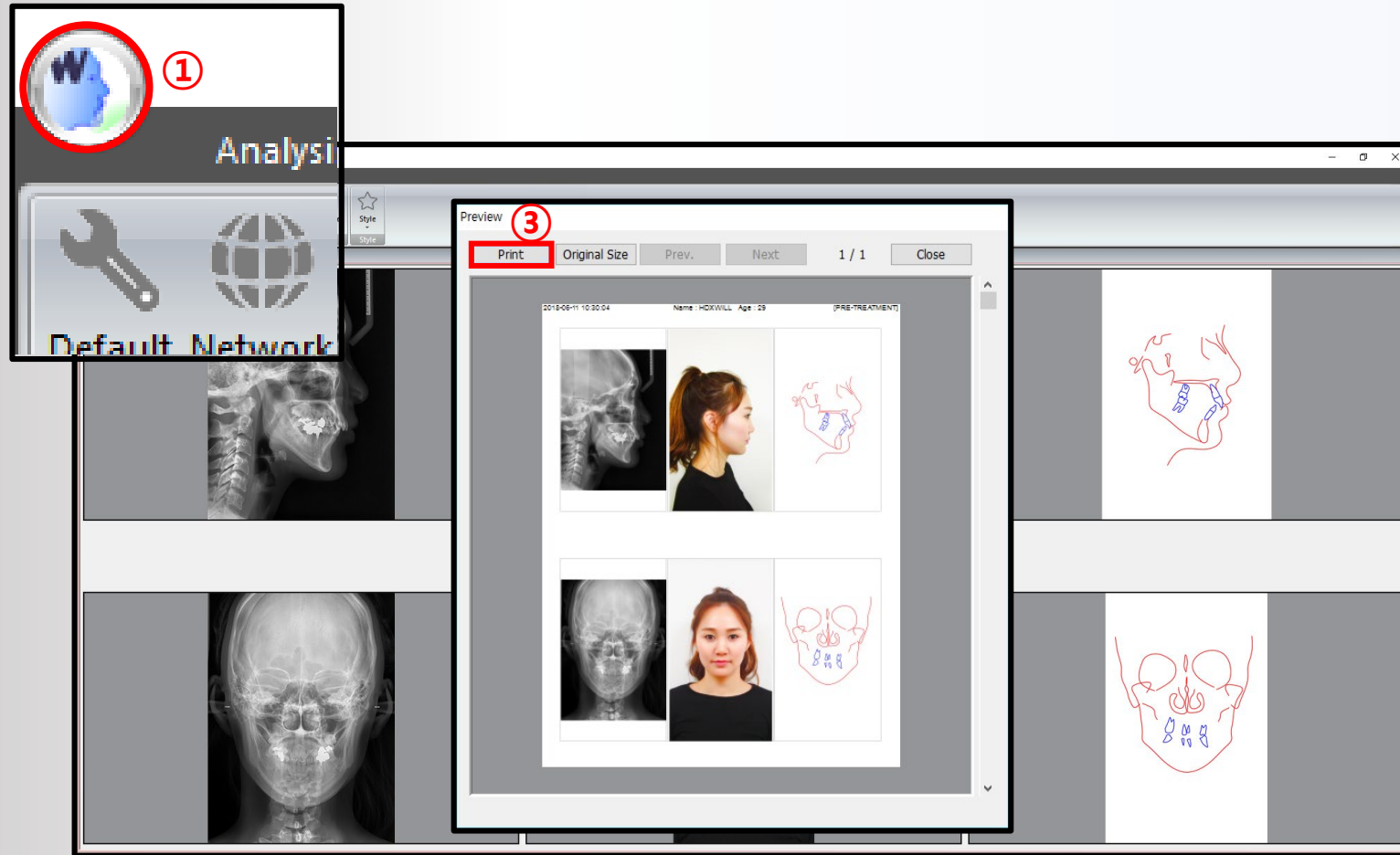
Gallery



1. Gallery Layout

- ① Gallery Layout
- ② Double click on any image to show it up on full screen (Hit Esc button of keyboard to exit the screen)

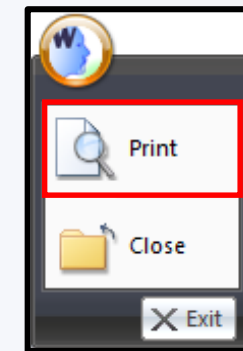
5) Gallery



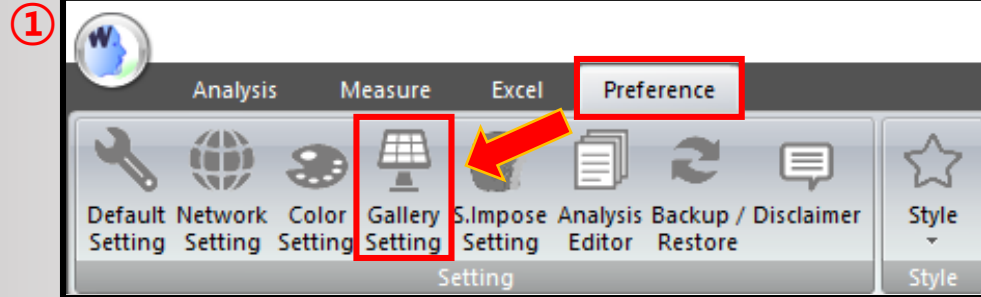
2. Print Gallery Layout

① Click  at the right side of the top

② Click [Print]



③ Click [Print] on the [Preview] window



3. Layout Setting – Layout selection

① Go to [Gallery Setting] under [Preference]

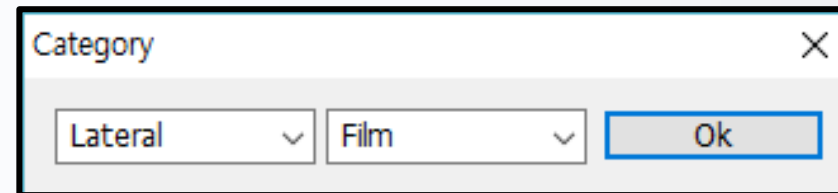
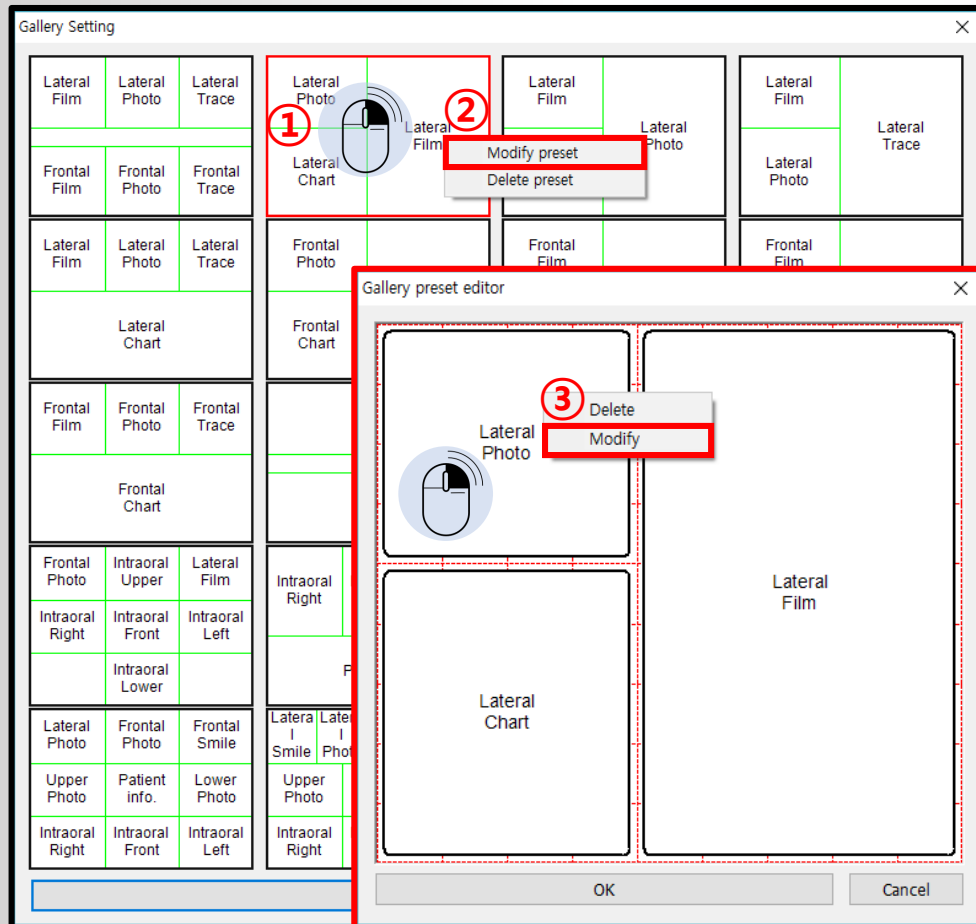
② Select preferred layout and click [Apply]



5) Gallery

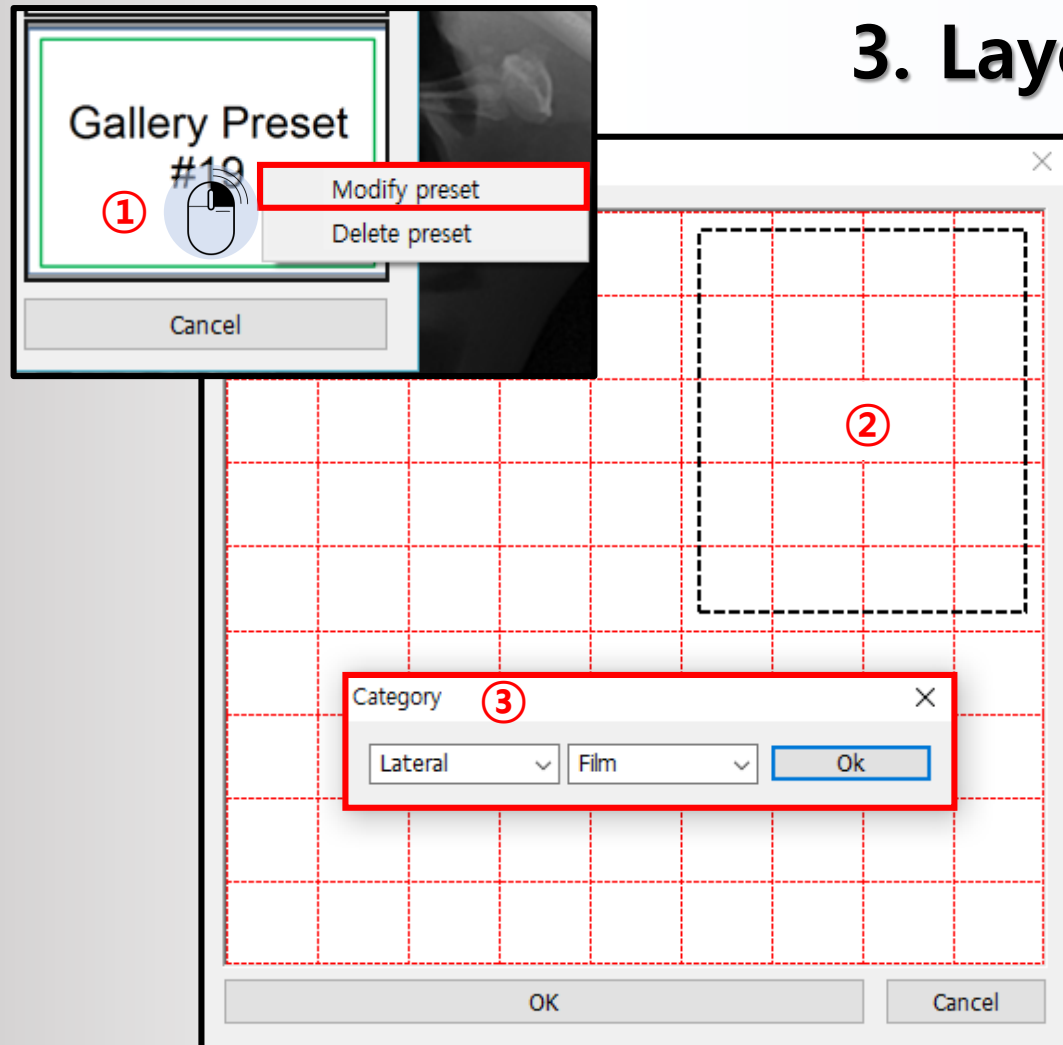
3. Layout Setting & Modification

- ① Right click on a layout which needs to be modified
- ② Click [Modify preset]
- ③ When [Gallery preset editor] window appears, right click on a spot that you want to modify and click [Modify]
- ④ Set the category of the spot

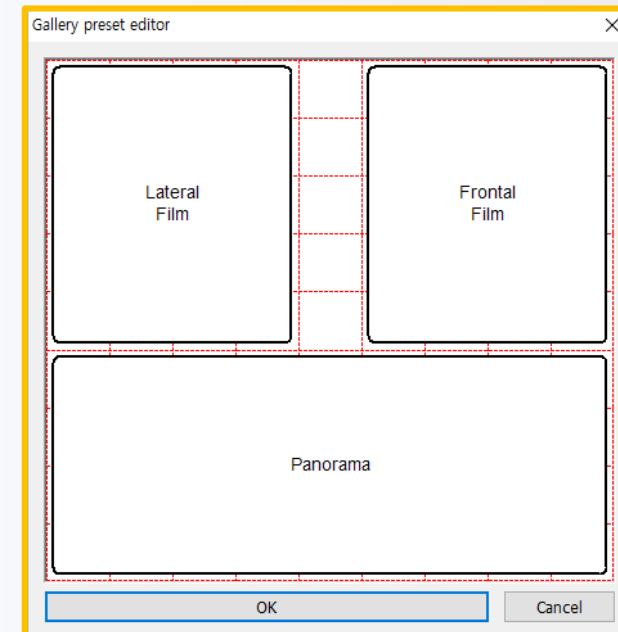


5) Gallery

3. Layout Setting – Personal Layout Creation



- ① Right click on the undesired layout and click [Modify preset]
- ② You can modify and adjust the area
- ③ Set category
(Example)



Thank
you



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Education & Development dept.